MIL-G-23081C(AS)
2 October, 1972
Superseding
MIL-G-23081B(AS)
13 January 1967

MILITARY SPECIFICATION

GYROSCOPE INDICATOR SYSTEM, VERTICAL

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification is a Vertical Gyroscope Indicator System operated by 115/200 volt, 400 Hertz, 3-phase, 4-wire wye power supply.

1.2 Classification - The Vertical Gyroscope Indicator System shall consist of the items listed in 6.7. The Gyroscope System shall be of the following types:

Type I	Three-Inch System
Type II	Five-Inch System

2. APPLICABLE DOCUMENTS

* 2.1 General - The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-5015	Connectors, Electric, AN Type
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For

FSC 6610

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MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-L-6085	Lubricating Oil, Aircraft Instrument, Low Volatility
MIL-M-7793	Meter, Time Totalizing
MIL-C-14806	Coating Reflection Reducing for Instrument Cover Glasses and Light Wedges
MIL-T-18303	Test Procedures: Preproduction and Acceptance For Aircraft Electronic Equipment, Format For
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-G-23083	Gyroscope, Rate Switching MS17399
MIL-L-25467	Light, Integral, Aircraft Instrument, General Specification For
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect
MIL-G-81322	Lubricating Grease, Ball Bearing
MIL-G-81559	Gyroscopic Assemblies, Attitude and Directional Reference Instrument For Aircraft Preservation and Packaging, Procedures For

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MIL-B-81744	Barrier Film Solvents and Solutions
Federal	
BB-N-411	Nitrogen
DD-G-451	Glasses, Flat and Corrugated, for Glazing, Mirrors and Other Uses
STANDARDS	
Federal	
FED-STD-209	Clean Room and Work Station Requirements, Controlled Environment
FED-STD-595	Colors
Military	
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipments
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-1334	Barrier Coating Process for Anti-Friction Bearing
MS3106	Connector, Plug, Electric, Straight
MS3116	Connectors, Plug, Electric Solder Type, Straight Bayonet Coupling
MS17322	Meter, Time Totalizing Miniature, Digital 115 Volt 400 Cycle
MS14114	Gyroscope, Displacement, Roll and Pitch Hardmount
MS17399	Gyroscope, Rate Switching
MS17400	Indicator, Attitude, Remote - 3-inch Case

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MS17401	Indicator,Attitude, Remote-5-inch Cas
MS24367	Lamp, Incandescent, Miniature Integral Lighting
MS25237	Lamp, Incandescent, Single Contact Midget Flanged Base
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for
AR-34	Failure Classification for Reliability Testing, General Requirements for
AS-2844	Lubricant, Instrument Ball Bearing High Flash Point

2.2 Availability of Documents - When requesting specifications standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center (1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials requirements shall conform to MIL-E-5400.

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(2) Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included, therein, shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4.)

(3) When previously produced models of this equipment do not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

* 3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including tubes, transistors and diodes) shall be obtained as outlined in MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

* 3.2.1.1 Final approval of all nonstandard parts and materials shall be completed before Preproduction approval can be granted.

** 3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5. Conformal coatings encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

* 3.2.3 Aluminum Alloy Parts - Aluminum parts shall be covered with an anodic conforming to MIL-E-5400.

* 3.2.4 Nonmagnetic Materials - Nonmagnetic materials shall be used for all parts of the Indicators, except where nonmagnetic materials are essential.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of MIL-E-5400 for design, construction, and workmanship, except as otherwise specified herein.

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* 3.3.1 Total Weight - The total weight of the system excluding cables, shall be as follows:

	Type I 3-Inch System	Type II 5-Inch System
Displacement Gyroscope with Mounting Base	7.0 lbs.	7.0 lbs.
Remote Attitude Indicator	3.8 lbs.	5.0 lbs.
Rate Switching Gyroscope	1.6 lbs.	1.6 lbs.
	12.4 lbs.	13.6 lbs.

* 3.3.2 Reliability - The contractor shall conduct a Reliability Program using MIL-STD-785. On a reorder from a supplier, who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

* 3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of at least 2000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

* 3.3.2.3 Reliability in Mean-Time-Between-Failure (MTBF) - The Gyroscope and Indicator shall each have an MTBF of 2000 hours. The MTBF of the Rate Switching Gyroscope shall be in accordance with MIL-G-23083.

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3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The system shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnection Cabling - The system shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088 and Figure 1 or Figure 2 of this specification. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated, as necessary, to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the system.

3.3.4 Interchangeability of Reordered Equipment - For reordered equipment, interchangeability shall exist between units and between all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment or system supplied or designated by the procuring activity.

* 3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461.

* 3.3.6 Provisions for Maintainability - The Maintainability Program, built-in test features, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in Requirement AR-10.

3.3.7 Nameplates - Nameplates approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory Bench Tests.

Temperature	Room ambient (25±5°C)
Altitude	Normal ground (0-5000 ft.)
Vibration	None
Humdity	Room ambient up to 90 percent relative humidity
Input power voltage	115 ± 1.0 VAC, 400 ± 5 Hz 115/200 ± 1.0 VAC 3 Phase 400 ± 5 Hz

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3.3.9 Service Conditions - The system shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 Equipment, except as modified herein.

* 3.3.9.1 Vibration - The Vertical Gyroscope system shall operate satisfactorily and no damage to any parts shall result when subjected to the Vibration requirements of Paragraph 4.8.41.

3.3.10 Primary Input Power Requirements - The system shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics, as defined in MIL-STD-704, having limits as specified therein. The power required shall not exceed the amounts specified.

(1) Displacement Gyroscope
AC Power (Three Phase) 115/200 V, Category "B",
55 VA Starting, 35 VA Running, and 55 VA Fast Erection

(2) 3-Inch Remote Attitude Indicator
AC Power (Single Phase) 115 V, Category "B", 22 VA

(3) 5-Inch Remote Attitude Indicator
AC Power (Single Phase) 115 V, Category "B", 22 VA

(4) Rate Switching Gyroscope
AC Power (Single Phase) 115 V, Category "B", 10 VA

3.3.10.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal Electric System Operation. Operation shall return to normal with no resulting damage to the equipment.

* 3.3.11 Hermetic Sealing - The case of each system component shall be hermetically sealed. The Electronic portion of the Displacement Gyroscope does not require Hermetically Sealed Design, except that it shall be gasket sealed.

3.3.11.1 Filling Medium- The filling medium of the Displacement Gyroscope shall be 95% helium and 5% carbon dioxide. The filling medium of the Attitude Indicators shall be 98% nitrogen and the remainder helium. The nitrogen used shall be of a quality in accordance with Specification BB-N-411B-1, Type I, Class 1, Grade A. The gases used shall be of at least 98% purity, free of dust particles and containing not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The absolute pressure of the filling medium shall be 1 $\pm$ 0.1 atmosphere.

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3.3.12 Wiring - The Internal Electrical Wiring of each system component shall be insulated from its case. The External Wiring shall be in accordance with Figure 1 for the Type I (3-Inch System) and Figure 2 for the Type II (5-Inch System).

3.3.13 Insulation Resistance - The Insulation Resistance of each system component must be 50 megohms or more when 500 Volts DC is applied between isolated electrical circuits and between isolated electrical circuits and the chassis.

3.3.14 Degree of Cleanliness - The equipment shall be assembled in a Class 100 clean room as specified in FED-STD-209.

* 3.3.15 Lubrication - The gimbal bearings shall be lubricated with oil, AS-2844, in accordance with MIL-I-6085. The spin motor bearings shall be lubricated with grease MIL-G-81322 or Minipur (or equivalent) with particle size not exceeding 25 microns for two percent and five microns for 98 percent of volume. Spin motors shall have porous separators impregnated with oil, AS-2844.

* 3.3.16 Barrier Coated Bearings - All Ball Bearings and Interfacing Associated Parts which may bridge and cause migration of lubrications shall be barrier coated with MIL-B-81744 in accordance with MIL-STD-1334. All ball bearings shall be ABEC 7 or better. Identification marking shall comply with MIL-N-18307.

* 3.3.17 Encapsulated Brushes - Encapsulated Brush and Slip Ring Assemblies, with optinum preset brush tension, shall be installed on the outer gimbal axis to prevent contamination and Gyroscope malfunction. Lead wires shall be stranded and slip rings shall have minimum run out. Electro-deposited gold 99.5% composition shall be used for slip rings.

3.3.18 Time Totalizing Meter - The system components, except the Rate Switching Gyroscope, shall contain time totalizing meters (10,000 hours) in accordance with MIL-M-7793 and MS17322. The reading of the meters shall not require unsealing of the components. The time totalizers shall be installed within the present envelope of the components.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The Vertical Gyroscope Indicator System shall provide gyroscopic stabilized Roll and Pitch Reference Data for aircraft utilization. The data shall be standard synchro-type signals, as specified herein.

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3.4.1.1 Attitude - The system shall provide aircraft roll and pitch outputs which are referenced to gravity. The outputs shall be gyroscope stabilized and the data shall be visually presented.

* 3.4.1.2 Output Accuracies - The system, when connected and operated as a system at standard conditions, shall be capable of providing the following:

Output	Accuracy
Roll and Pitch	0.25 degree max.
System Drift	0.25 degree/min. max.

3.4.1.3 Performance at Environmental Extremes - Performance deviations at Environmental Extremes shall be permitted as specified in the detail requirements.

3.4.1.4 System Interconnections - The system shall provide signals and control circuits as shown in Figure 1 for Type I and Figure 2 for Type II. The Interconnecting Cabling shall not be supplied as part of the system.

3.4.3 Accelerations

* 3.4.3.1 Longitudinal Acceleration - The system, when subjected to an 8g acceleration for 4 seconds along the roll axis, shall provide a pitch and roll error not to exceed 2° maximum.

* 3.4.3.2 Coordinated Turns - The system, when subjected to coordinated turns with rates of turn between 90 and 1500 degrees per minute and bank angles of 20 to 60 degrees, shall provide a pitch and roll error not to exceed 2° maximum after 8 minutes.

* 3.4.3.3 Uncoordinated Turns - The system, when subjected to uncoordinated turns (skid turns) of less than 5 degrees per minute at air speeds not greater than 400 knots for a 10-minute duration, shall provide a pitch and roll error not to exceed 20 maximum.

3.5 Detail Requirements -

3.5.1 Displacement Gyroscope - The reference for the Vertical Gyroscope Indicator System is the Displacement Gyroscope. The Displacement Gyroscope shall be constructed in accordance with MS14114 and the requirements of this specification. It shall develop continuous electrical signals representative of aircraft attitude through 360 degrees roll and ± 82 degrees of pitch. The Displacement Gyroscope shall operate with all the components of the Vertical Gyroscope Indicator System. It shall be built to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service.

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*3.5.1.1 Case - The outline dimensions of the Displacement Gyroscope and its shockmount shall be in accordance with MS14114-2.

*3.5.1.2 Gyroscopes Mount - The Gyroscope shall have internal vibration isolators and shall be hard-mounted in accordance with MS14114-3.

* 3.5.1.2.1 Cable and Connector - With the Displacement Gyroscope properly mounted on the Gyroscope Mount, the assembly shall meet the requirements of MS14114.

* 3.5.1.2.2 Attaching Method - The Gyroscope Mount shall incorporate a means of rapidly clamping the feet of the Displacement Gyroscope in place. One end of the mount shall hold the Gyroscope firmly by the feet when the case is placed on the mount and slid along its long axis toward that end for a distance not greater than 5/8 inch. The other end of the mount shall permit the quick release of the Displacement Gyroscope Case from the mount by means of a thumbscrew quick release mechanism on the mount. Normally the cable end of the Displacement Gyroscope Case shall be on the quick release end of the mount. Replacement of the Gyro shall not require realignment in the airframe.

* 3.5.1.3 Gimbaling - The Displacement Gyroscope shall be mounted in two gimbals and shall have two axes of freedom. There shall be a synchro and torque motor on each axis. The pitch axis shall be the pivot axis of the inner gimbal (gyro wheel housing) and shall be perpendicular to the roll axis which is the pivot axis of the outer gimbal. The roll axis shall be parallel to the longitudinal axis or long centerline of the case as shown on MS14114.

3.5.1.4 Gimbal Freedom - The Displacement Gyroscope shall have unlimited freedom in roll and a minimum of plus or minus 82 degrees of freedom in pitch. Pitch stops shall be incorporated which produce a roll precession when the pitch freedom is exceeded. The pitch stops shall be so designed that no parts of the Gyroscope will be damaged by a loop at the rate of 90 degrees per second. The plus or minus three degrees error shall result from loops of 15 degrees per second plus or minus one degree per second. Errors induced by roll maneuvers at rates plus or minus three degrees of error to not exceed one degree.

3.5.1.5 Drift - Unless otherwise specified, the Displacement Gyroscope shall be so designed that the maximum free drift, noted after compensation for the earth's rotation, shall not be greater than 0.25 degree per minute. This requirement applies over the entire temperature range specified herein, and with the spin axis either tilted 20 degrees or vertical.

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3.5.1.6 Erection - The erection mechanism shall incorporate separate systems for pitch and roll. Erection shall be accomplished by electrolytic switches and torque motors on the gimbals. When the vertical is displaced in one axis, it shall not deviate by more than one degree in the other axis while it is erecting. The erection system shall provide both Initial and Final Erection and shall erect the Displacement Gyroscope, from any attitude to the vertical, as determined by the Vertical Sensing Electrolytic Switches. The above shall also be accomplished with the Gyroscope hardmounted and without Vibration.

3.5.1.6.1 Initial Erection - The Displacement Gyroscope shall have fast Initial Erection such that it assumes a position within one degree of vertical in pitch and roll within a maximum of 68 seconds after power is applied.

3.5.1.6.2 Final Erection - The Displacement Gyroscope shall align its spin axis with the direction of gravity within plus or minus 1/2 degree in two minutes, or less, after the Initial Erection period.

3.5.1.6.3 Erection Rate -

3.5.1.6.3.1 High (Fast) Erection - Means shall be provided to apply a fast erection simultaneously to both axes of the Gyroscope to the apparent vertical. This fast erection shall be obtained by externally closing the circuit between Pins N and C of the Electrical Connector. The fast erection rate shall not be less than 15 degrees per minute in either axis.

3.5.1.6.3.2 Low (Normal) Erection - The rate of normal erection about the roll axis shall be nominally two degree per minute and shall fall between the limits plotted on Figure 3 throughout all environments and all roll attitudes of the Displacement Gyroscope Case. Roll erection shall not be affected appreciably by displacement of the Gyroscope from vertical about the pitch axis through plus or minus 82 degrees. The rate of normal erection about the pitch axis shall be nominally one degree per minute and shall fall between the limits plotted on Figure 3 throughout all environments and pitch attitudes of plus or minus 40 degrees. For greater pitch attitudes, the erection rate shall be held to a minimum but shall not be less than that permitted by Figure 3. Pitch erection shall not be affected appreciably by displacements of the Gyroscope in the roll axis of plus or minus 85 degrees or less. Internal provisions shall also be made for Internal Roll Cut-out without change to existing connector.

3.5.1.6.3.3 Roll and Pitch Erection Cutout - The design shall be such that when Pins L and G are connected externally, roll erection shall be cut out and when Pins K and G are connected externally, pitch erection shall be cut out. The current drawn from Pin G to each pin (L and K) shall not exceed 250 ma. Internal provisions shall also be made for internal roll cut out without change to the external connector.

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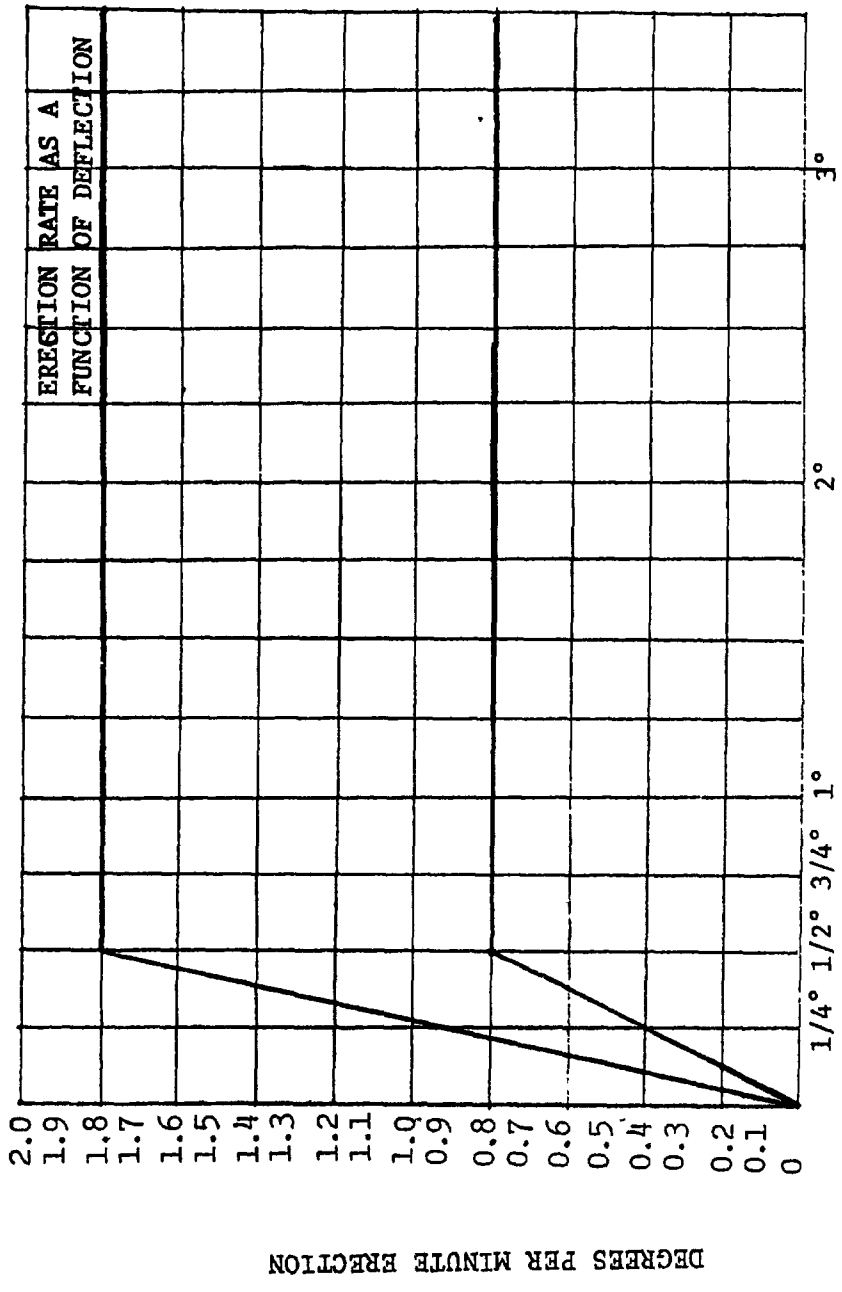


FIGURE 3. LOW (NORMAL) ERECTION RATE, DISPLACEMENT GYROSCOPE

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* 3.5.1.7 Pitch and Roll Outputs - The pitch and roll outputs shall be of a standard synchro type with a no load output of 11.8 ± 1/4 volt line-to-line and a no-load phase shift which shall not exceed 5.5 degrees. The roll output shall be continuous through 360 degrees and the pitch output shall be continuous through plus or minus 82 degrees. Each synchro shall follow its respective gimbal movement to a one-to-one ratio. The Displacement Gyroscope shall have brushes and slip rings providing maximum redundant circuits to insure continuity of signal outputs. The Gyroscope and Brush Assembly shall be capable of withstanding 700 volt transients of 25 milliseconds duration. The Pitch Synchro output shall not show effects of cross coupling due to Aircraft Pitch or Roll maneuvers greater than plus one minus zero degrees for roll attitude up to 30 degrees.

3.5.1.7.1 Synchro Transmitter Characteristics - Each transmitter shall be capable of operating up to five Bendix, Type AY500-5, (or equivalent) Control Transformers. When loaded with five loads, the electrical characteristics of each transmitter shall not vary by more than the following:

(1) The voltage drop from the nominal unloaded line-to-line output shall not exceed 0.6 volt with the rotor leads of each control transformer open.

(2) The angle measured between the exciting voltage of each transmitter rotor and the voltage in any of the three stator leads of each of the five control transformers shall not exceed 3.5 degrees. This measurement shall be made with the rotor leads of the five control transformers open.

(3) Nulls measured with up to five loads applied shall not exceed 0.03 volt.

The conditions specified in (1) and (2) shall apply to the pitch and roll transmitters at room temperature.

* 3.5.1.7.2 Electrical Connector - The Electrical connectors shall conform to MIL-C-5015 and shall be of the type specified in MS14114. The connector shall be affixed to the Displacement Gyroscope as shown on MS14114 with pin connections as shown in Table I. The key way on the connector shall be located in the normal zero degree position with respect to the pin insert. All electrical circuits shall be isolated from the case. The connector shall mate with MS3106-22-14S.

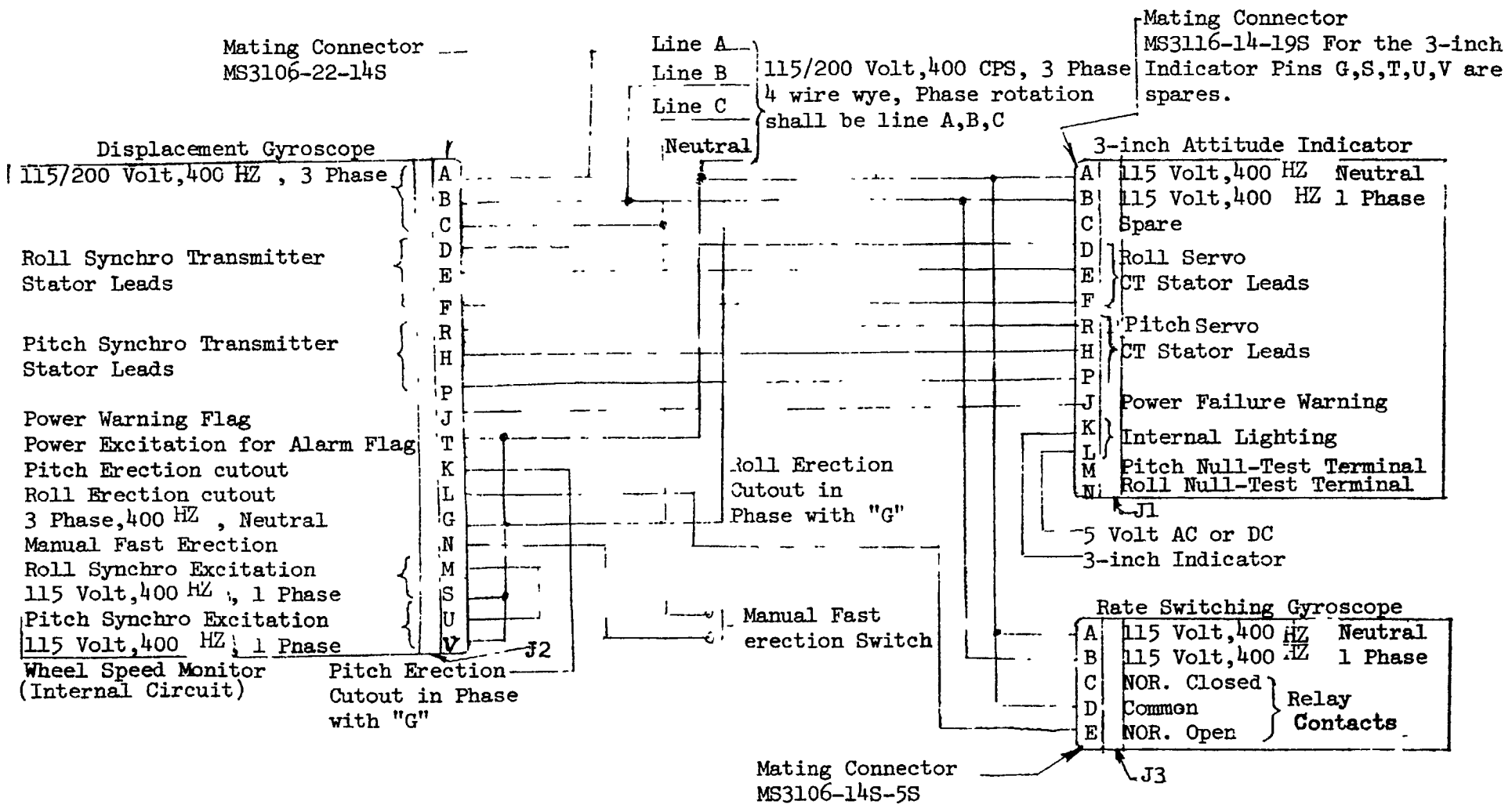
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TABLE I

ELECTRICAL CONNECTOR PIN CONNECTIONS, DISPLACEMENT GYROSCOPE

J2 Pins	Circuit
A	115/200V, 400 Hz A — Phase rotation
B	115/200V, 400 Hz B — A, B, C
C	115/200V, 400 Hz C
D (x)	Roll transmitter synchro stator
E (y)	
F (z)	
R (x)	Pitch transmitter synchro stator
H (y)	
P (z)	
G	3 Phase, 400 Hz neutral
J	Power warning flag
T	Power excitation for alarm flag
K	Pitch erection cutout
L	Roll erection cutout
M (H)	Roll synchro excitation 115V, 400 Hz ϕ B
S (C)	
U (H)	Pitch synchro excitation 115V, 400 Hz ϕ B
V (C)	
N	Manual fast erection

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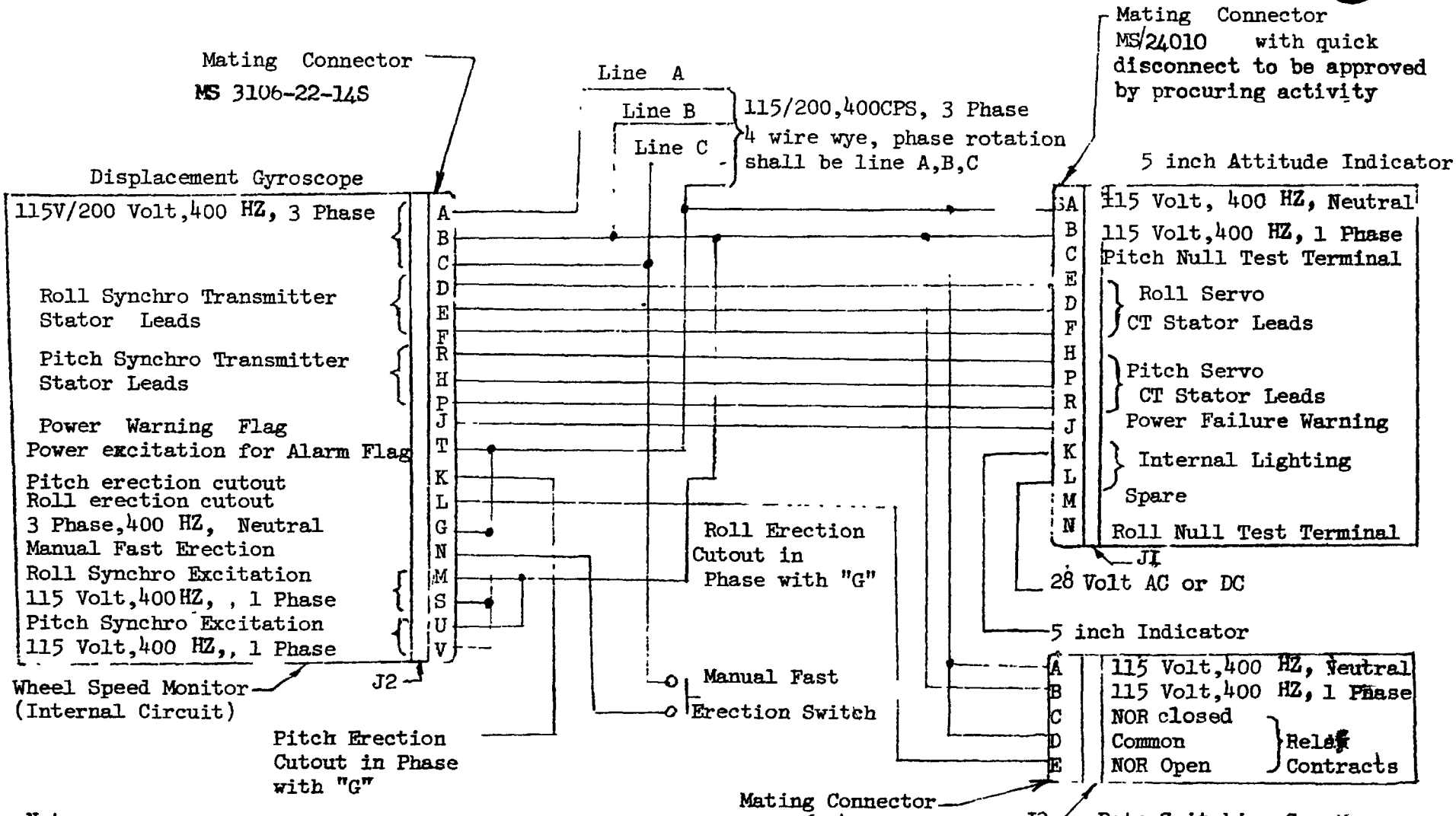
NOTES:

1. Wires D,E,F, and R,H,P are to be twisted as grouped with at least six turns per foot.
2. Connect "K" on gyroscope for pitch erection cutout from fore/aft acceleration sensing mechanism to 115/200 V. Neutral during acceleration.
3. Connect "L" on gyroscope for roll erection cutout from rate gyroscope turn sensing mechanism to 115/200 V, 400 HZ Neutral during turns.
4. For manual fast erection, 115 Volt Phase A, B, or C should be connected to pin "N" on gyroscope

FIGURE 1: SYSTEM INTERCONNECTING DIAGRAM, 3-INCH INDICATOR

MIL-G-23081C

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Notes:

1. Wires D,E,F and R,H,P are to be twisted as grouped with at least six turns per foot.
2. Connect "K" on gyroscope for pitch erection cutout from fore/aft acceleration sensing mechanism to 115/200 V, Neutral during acceleration.
3. Connect "L" on gyroscope for roll erection cutout from rate gyroscope turn sensing mechanism to 115/200 V, 400HZ, neutral during turns.
4. For manual fast erection, 115 volt phase A,B or C should be connected to pin "N" on gyroscope.

FIGURE 2. SYSTEM INTERCONNECTING DIAGRAM, 5-INCH INDICATOR

MIL-G-23081C

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3.5.1.8 Synchro Zeroing and Sensing - The synchros shall be zeroed and their lead terminals shown in Table I connected for the following procedure:

(1) Connect Terminals S and V to Pin G (neutral) of the three phase 115/200 volts 400 Hz power source.

(2) Connect Terminals M and U to Pin B (115 Volt phase B) of the power source.

(3) Connect Terminals F and P to Pin G (commoned point of S and V) with the Gyroscope gimbals at zero displacement about their axes. The voltage between D and E and between R and H shall be at a null.

(4) With D and E strapped, the voltage between this point and M shall be less than 115 volts.

(5) With R and H strapped, the voltage between this and U shall be less than 115 volts.

(6) With D and E unstrapped and the Gyroscope Case displaced in right-bank about the roll axis with respect to the roll gimbal, the voltage between D and Pin G (the commoned point of V and S) shall increase before decreasing.

(7) With R and H unstrapped and the Gyroscope Case displaced in climb about the pitch axis respect to the pitch gimbal, the voltage between R and Pin G (the commoned point of P and V) shall increase before decreasing.

3.5.1.8.1 The synchro transmitter leads are defined with reference to the external electrical receptacle pins for the above procedure in Table I.

* 3.5.1.9 Power Interruption - Means shall be provided in the Displacement Gyroscope to prevent system recycling through initial erection when power has been removed. After five minutes of operation at normal voltage, the Gyroscope shall not recycle or develop errors greater than plus or minus one degree for 30 seconds after power has been removed. Gyroscope Gimbal Freedom shall not be less than plus or minus 82 degrees in pitch and plus or minus 85 degrees in roll when power has been removed.

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3.5.1.10 Power Failure Warning - A 3-phase power warning circuit shall be incorporated in the Displacement Gyroscope. This circuit shall operate a set of normally open contacts. Leads from this set of contacts shall be brought out of the case to Pins J and T as shown in Table I. When proper power is applied to the Gyroscope the contacts will close the circuit between Pins J and T, after the intial erection cycle is completed. If at any time any phase of the 3-phase power to the Gyroscope is broken, the contact between Pins J and T will break.

* 3.5.1.10.1 Gyro Wheel Speed Monitor - A Gyro Wheel speed monitor circuit shall provide a power failure flag warning when the Gyro Wheel speed drops below 5000 RPM.

3.5.2 Attitude Indicator - The Attitude Indicator shall be capable of presenting pictorial display of the aircraft pitch and roll attitude throughout all maneuvers, correlated with scalar indexing of mechanisms for actuating pitch and roll gimbals, a dial display consisting basically of a fixed miniature airplane, a bank angle dial, a bank index, a moving two-colored spheroid background with a horizontal line dividing the colors, a pitch trim knob and mechanism, an external adjustment for pitch and roll, and a power failure indicating device. The Type II indicator (5-Inch) shall include a mechanism mask. The indicator shall be so designed that the complete mechanism can be removed from the case as a unit. It shall be built to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, and installation.

3.5.2.1 Indicator Display - The display shall consist basically of a fixed miniature airplane, a bank angle dial, a bank index, and a moving two-colored spheroid background with a distinct horizon line dividing the two colors. All colors shall be in accordance with FED-STD-595, unless otherwise specified. The finish shall be such that it diffuses light sufficiently to avoid "high-light" when illuminated with external or internal light sources.

3.5.2.1.1 Miniature Airplane -

3.5.2.1.1.1 Type I (3-Inch) - The miniature airplane shall be in accordance with Figure 4. The rear edge of the miniature airplane shall be within 1/16 inch of the spheroid surface. The wings shall follow the curvature of the spheroid. The top edge of the wings and center dot shall be tilted back approximately 15 degrees to facilitate night lighting. The miniature airplane shall be fixed to the bank angle dial. The "W" on the miniature airplane shall conform with Figure 4, the white portion of the "W" and the wings and dot must be separated by an area of black Color No. 37038 of sufficient width to give an area of contrast. The white areas of the miniature airplane shall be Color No. 37875.

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* 3.5.2.1.1.2 Type II (5-Inch) - The miniature airplane shall be in accordance with Figure 6. The rear edge of the miniature airplane shall be within 1/16 inch of the spheroid surface. The wings shall follow the curvature of the spheroid. The top edge of the wings and center dot shall be tilted approximately 15 degrees to facilitate night lighting. The miniature airplane shall be fixed to the bank angle Dial. The attaching areas shall not exceed 1/8 inch diameter each and shall be black (Color No. 37038) of FED-STD-595. The "W" on the miniature airplane need not have a black border as shown on Figure 6; however, the white portion of the "W", wings, and dot must be separated by an area of black (Color No. 37038) of sufficient width to give an area of contrast. This shall be subject to the approval of the procuring activity. The white areas of the miniature airplane shall be white (Color No. 37875).

3.5.2.1.2 Mechanism Mask - A mask with the visible surface shaped as the surface of the frustrum of a cone shall be attached to the roll gimbal and shall extend from the bank angle dial and shall be approximately 3-11/16 inches on the large diameter. This edge of the mask shall be hidden by the inner diameter of the bank angle dial. The smaller diameter of the mask shall be closest to the spheroid and shall intersect the spheroid in such a manner that a lateral opening of at least 3-3/8 inches results. This edge of the mask shall conform to the shape of the spheroid to within 1/16 $\pm$ 1/32 inch.

* 3.5.2.1.3 Bank Angle Dial - The bank angle dial shall be in accordance with Figure 4 for Type I (3-Inch) and Figure 6 for Type II (5-Inch). It shall be affixed to, and shall rotate with, the roll gimbal but shall not move with pitch deviations. The front surface of the bank index shall be in a plane parallel to the cover glass. This plane shall be in the same plane as the front surface of the small diameter of the bank dial within plus or minus 3/64 inch. The arc described by the small end of the bank index shall not deviate more than 3/64 inch radially from the calibrated small diameter portion of the bank dial. The index shall be white (Color No. 37875) with a uniform black border and black support (Color No. 37038).

3.5.2.1.4 Spheroid - The spheroid shall be mounted in the roll gimbal and shall rotate with the gimbal about the roll axis for bank indications and it shall rotate in relation to the roll gimbal about the pitch axis for pitch indications. The roll axis shall be perpendicular to the face of the Attitude Indicator, and the pitch axis shall be horizontal when the Indicator is level and indicating a zero roll attitude. The horizon line on the spheroid shall be fixed to the spheroid at the zero pitch position. The spheroid shall be driven in pitch so that during a dive

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maneuver the horizon line goes above the miniature airplane, and during a climb maneuver the horizon line goes below the miniature airplane. The spheroid shall be in accordance with Figure 5 for Type I (3-Inch) and Figure 7 for Type II (5-Inch). The surface of the spheroid shall move 0.035 (nominal) inch for each degree of pitch change. The colors and markings shall be in accordance with Figure 5 for Type I (3-Inch) and Figure 7 for Type II (5-Inch) and the following requirements. The spheroid shall be internally illuminated.

3.5.2.1.4.1 Colors and Markings - Above the horizon line, the spheroid shall be gray (Color No. 36440) and all markings shall be black (Color No. 37038 of FED-STD-595). The horizon line shall be white (Color No. 37875). Below the horizon line the spheroid shall be black (Color No. 37038) and all markings shall be gray (Color No. 36440). The lengths of the major pitch lines shall be as follows:

	For Type II (5-Inch)	For Type I (3-Inch)
0° (Horizon line) - - - - - - - - -	Width of spheroid	Width of spheroid
10° - - - - - - - - - - - - - - - -	2-1/2 inches	1-11/16 inches
20° - - - - - - - - - - - - - - - -	2-3/8 inches	1-17/32 inches
30° - - - - - - - - - - - - - - - -	2-1/4 inches	1-11/32 inches
40° - - - - - - - - - - - - - - - -	2-1/8 inches	1-5/32 inches
50° - - - - - - - - - - - - - - - -	2 inches	31/32 inch
60° - - - - - - - - - - - - - - - -	1-7/8 inches	25/32 inch
70° - - - - - - - - - - - - - - - -	1-3/4 inches	19/32 inch
80° - - - - - - - - - - - - - - - -	Circle .720 Dia.	Circle 7/16 inch dia
85° - - - - - - - - - - - - - - - -	Solid dot .360 Dia.	Solid dot 1/4 inch dia

The center of the solid dots shall be the plus or minus 90-degree pitch position and from center to center shall represent 180 degrees of pitch indication. For Type I (3-Inch) the pitch indicator shall be linear over the full range and this 180 degree range shall be spread over 270 degrees of the spheroid. For Type II (5-Inch) the pitch indication shall be linear ov the full range and this 180-degree range shall be spaced over 180 degrees of the spheroid surface. All markings shall be durable to withstand usage encountered in service. The form of the letters and numerals shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies.

* 3.5.2.2 Attitude Actuation - The spheroid shall be rotated about the pitch and roll gimbal axes by means of geared servo motors until a null is attained between the synchro control device on the Attitude Indicator axes and the transmitter synchros in the pitch and roll displacement reference The freedom of the spheroid rotation shall be unlimited about the roll axes. Stops shall be incorporated in pitch such that the indication will not become unsynchronized with the input signal, nor the indicator become damaged when th input pitch signal exceeds the plus or minus 90 degree pitch range of the Indicator and comes within the range again by continued rotation in the same direction. Adjustable anti-backlash gears or other means shall be used when required to prevent spheroid jitter resulting from vibrations.

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* 3.5.2.2.1 Indicator Synchro Control Device - The synchro control devices on the pitch and roll axes shall be high impedance synchros so that when one to four indicators are connected to a single test transmitter per axis, no noticeable change in performance can be noted.

3.5.2.2.2 Standard Test Transmitter - The Standard Test Transmitter shall consist of an Eclipse-Pioneer Type AY200-1, or equal, high precision synchro with minimum and known errors. The rotor of the synchro shall be settable and rotatable through 360 degrees. These settings shall be readable to within six minutes of arc and each degree of scale marking shall represent one degree of synchro rotor rotation. The following method will be used to calibrate the standard test transmitter:

(1) The rotor leads shall be designated as H and C and one stator lead designated as Z and connected to C. When 26 volts is applied across H and C (C being ground) the scale shall be positioned on zero when the voltage across H and X is maximum and the voltage across the remaining two stator leads (X and Y) is minimum. The voltage across X and Z increases before it decreases for increasing numbers of the scale.

(2) Synchro Excitation - The test transmitter rotor shall be excited from the same power supply voltage that operates the Attitude Indicator. It shall be so applied that the voltage from rotor lead C to rotor lead H is in phase with the voltage from Pin A to Pin B of the Indicator connector with lead C and Pin A common.

3.5.2.2.3 Pitch and Roll Sensing - When the standard test transmitters set at a scale reading of 180 degrees are connected to the Attitude Indicator as shown in Table II and III, the indication shall drive to zero pitch and roll. Increasing scale setting of the test transmitters shall result in right bank and climb indications. Decreasing scale setting shall result in left bank and dive indications.

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TABLE II

ROLL SERVO, ATTITUDE INDICATOR

Test Transmitter Terminal	Indicator Terminal of Power Input Connector		Power Excitation of Standard Test Transmitter
	For Type I (3-Inch)	For Type II (5-Inch)	
Hot (H)			26 V tap-off from single phase power supply to Pins A and B of indicator
Cold (C)			
X	D	E	
Y	E	D	
Z	F	F	

TABLE III

PITCH SERVO, ATTITUDE INDICATOR

Test Transmitter Terminal	Indicator Terminal of Power Input Connector		Rotor Excitation of Standard Test Transmitter
	For Type I (3-Inch)	For Type II (5-Inch)	
Hot (H)			26 V tap-off from single phase power supply to Pins A and B of indicator
Cold (C)			
X	R	H	
Y	H	P	
Z	P	R	

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3.5.2.2.4 Follow-up Operation - The electrical and mechanical design of the Attitude Indicator shall be such that in the process of follow-up, the indications in pitch or bank shall be smooth without hunting, oscillations, or jumping greater than 1/4 degree total amplitude in pitch or one degree total amplitude in bank when connected to a standard test transmitter. There shall be no sustained electrical or mechanical oscillations in either axis greater than 1/4 degree total amplitude for any indication. The servo mechanism shall be 70 to 100 percent damped and shall not lag the signal source by more than 10 degrees at maximum rate of servo operation.

* 3.5.2.2.5 Follow-up Rate - The rate at which the motors drive the gimbals shall be no less than 180 degrees per second in roll nor 90 degrees per second in pitch.

3.5.2.2.6 Follow-up Accuracy - The pitch and roll indications shall follow the input signals from their respective test transmitter synchro to within plus or minus 1/2 degree for positions plus or minus 30 degrees in roll and pitch, and 1.0 degree for all other positions.

3.5.2.2.7 Amplifier Channels - The Attitude Indicator shall incorporate rugged reliable transistor amplifiers with parallel or redundant circuits, insofar as practical, so the Indicator will continue to operate with reduced performance should partial failure of the amplifier occur.

3.5.2.2.8 Zeroing Adjustments - Means shall be incorporated to externally adjust the position of the spheroid plus or minus three degrees minimum in roll and plus or minus five degrees minimum in pitch. The initial setting of the pitch and roll indication at the time of delivery of the Attitude Indicator shall indicate zero pitch and roll, within plus or minus 1/4 degree when the Attitude Indicator is connected to a test transmitter set at electrical zero. The roll adjustment shall be on the rear of the case and shall be adequately marked. It shall be equipped with stops, so located and of sufficient strength to prevent inadvertent damage to the control.

3.5.2.2.9 Case - The outline dimensions of the Attitude Indicator Case shall conform to MS17400 for Type I (3-Inch) and MS17401 for Type II (5-Inch).

3.5.2.2.9.1 Body and Mounting Flange - The Body and Mounting Flange of the Attitude Indicator shall be made of lightweight nonmagnetic metal, uniform in texture, and shall have a smooth surface. The body shall be finished in a durable black finish and the mounting flange shall be finished with a durable lusterless black finish (Color No. 37038 of FED-STD-595).

3.5.2.2.9.2 Cover Glass - The cover glass shall conform to DD-G-451, Type II, Quality AA. Any flaws permitted by DD-G-451 shall not interfere with the reading of the Attitude Indicator. The edges of the glass shall be ground or otherwise treated to remove roughness and nicks. The thickness shall not exceed 0.205 inch.

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3.5.2.2.9.3 Dial to Cover Glass Distance - The distance of the front surface of the dial shall be approximately 0.375 inch for Type I (3-Inch) and 0.5 inch for Type II (5-Inch) from the inner surface of the cover glass.

* 3.5.2.2.9.4 Anti-Reflection Coating - A magnesium fluoride coating in accordance with MIL-C-14806 shall be applied to each surface of the cover glass and lighting wedge. Coated surfaces shall not chip, scale, peel, flake, dissolve, discolor, or otherwise be adversely affected by the Environmental Tests, specified herein, or in normal service use. The coating on the outer surface of the glass shall be capable of withstanding abrasion incident to normal service usage.

3.5.2.2.5 Electrical Power Variation - The Attitude Indicator shall operate properly and continuously for any combination of voltage and frequency as specified in MIL-STD-704, Category "B".

3.5.2.3 Pitch Trim - Means shall be incorporated to adjust the position of the spheroid with a pitch trim knob mounted on the lower right hand corner of the Attitude Indicator and marked in accordance with MS17400 for Type I (3-Inch) and MS17401 for Type II (5-Inch). The reference mark on the knob shall extend across the end of the knob and continue along the side to a point on the knob that is adjacent to the triangular mark on the bank angle dial at the zero pitch trim position for Type I (3-Inch) and adjacent to the circular mark on the case at the zero pitch trim position for Type II (5-Inch). These marks shall be white (Color No. 37875 of FED-STD-595). The knob shall have approximately one-half turn of freedom in the clockwise direction and one-quarter turn of freedom in the counter-clockwise direction from the zero pitch trim position. The freedom in each direction shall be limited by a positive stop. When the knob is rotated to the stop in the clockwise direction, the spheroid shall rotate to deflect the horizon line upward to indicate between 10 degrees and 20 degrees dive. When the knob is rotated to the stop in the counterclockwise direction, the spheroid shall rotate to deflect the horizon line downward to indicate between five degrees and 10 degrees climb. The servo system shall operate properly for any pitch trim adjustment. The knob shall have sufficient friction to remain fixed when set but it shall move smoothly without evidence of binding or erratic motion when being adjusted. The pitch trim circuit shall be such that in the event two Indicators are connected to one transmitter, varying the pitch trim setting on one of the Indicators shall not affect the indication of the other by more than 1/2 degree. The initial torque required to rotate the pitch trim knob from any setting shall be between the limits of two and six ounce-inches.

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* 3.5.2.4 Power Failure Indication - A Power Failure Warning Indication shall be incorporated in the lower left of the face of the Attitude Indicator on the bank angle dial as shown on MS17400 and Figure 4 for Type I (3-Inch) and MS17401 and Figure 6 for Type II (5-Inch). The word OFF shall be black on a white background. A lead shall be brought out in the electrical connector through Pin J, and the circuit shall be so designed that when this pin is connected to Pin A and power is applied the warning shall disappear When the connection is removed or the power is removed (or both) the warning shall appear. The warning shall appear due to electrical failure within the indicator, insofar as practical. The black letters shall be legible under night lighting conditions. The lighting shall not be entirely dependent upon one light source. The off flag voltage shall be VAC.

3.5.2.5 Electrical Connector -

3.5.2.5.1 Type I (3-Inch) - The Electrical Connector shall conform to MIL-C-26482 and shall be affixed to the Attitude Indicator Case as shown on MS17400, except that it may be in either of the lower corners of the case. This connector shall be a hermetically sealed receptacle and shall mate with MS3116-14-19S, or equivalent. The electrical connections shall be in accordance with Table IV.

* 3.5.2.5.2 Type II (5-Inch) - The Electrical Connector shall be Lear Part No. 110380-01 (or equivalent),and shall be affixed to the case as shown in MS17401, except that it may be placed in the lower right hand corner. The header and insert containing the connector pins shall be MS24010. The Mating Hood Assembly shall be approved by the procuring agency. The electrical connections shall be in accordance with Table V.

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TABLE IV

ELECTRICAL CONNECTOR PIN CONNECTIONS,
ATTITUDE INDICATOR TYPE I (3-Inch)

Electrical Pins	Connections
A	115 V 400 Hz neutral
B	115 V 400 Hz single-phase power (in phase with excitation to test transmitter synchros)
D (x) E (y) F (z)	The three stator leads for the roll synchro control device
R (x) H (y) P (z)	The three stator leads for the pitch synchro control device
J	Power failure warning (connect to Pin A to complete circuit)
K L	Internal lighting power input 5 volts (ac or dc)
M	Pitch null test terminal (pitch error voltage between this pin and Pin A)
N	Roll null test terminal (roll voltage between this pin and Pin A)
C G S T U V	No connection

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TABLE V

ELECTRICAL CONNECTOR PIN CONNECTIONS,
ATTITUDE INDICATOR TYPE II (5-Inch)

Electrical Pins	Connections
A	115 V 400 Hz neutral
B	115 V 400 Hz single-phase power (in phase with excitation to test transmitter synchros)
C	Pitch Null Test Terminal (Pitch error voltage between this pin and Pin A)
E (x) D (y) F (z)	The three stator leads for the roll synchro control device
H (x) P (y) R (z)	The three stator leads for the pitch synchro control device
J	Power failure warning (connect to Pin A to complete circuit)
K L	Internal lighting power input 28 volts (ac or dc)
M	No connection
N	Roll null test terminal (roll error voltage between this pin and Pin A)

3.5.2.6 Visibility of Display - The pitch scale graduations from 20 to -20 degrees, the bank index, the numerals, and other specified markings on the dial shall be visible from any point within the frustrum of a cone whose side makes an angle of 30 degrees with a perpendicular to the spheroid and whose small diameter is the aperture of the instrument case. Lamps conforming to MS24367 shall be used for Type I (3-Inch) and MS25237 for Type II (5-Inch).

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3.5.2.6.1 Illumination of Display - The Attitude Indicator shall be integrally lighted in accordance with the requirements of MIL-L-25467.

*3.5.2.6.2 Numerals and Letters - All numerals and letters on the Indicator shall be in accordance with MS33558.

3.5.3 Rate Switching Gyroscope - The Rate Switching Gyroscope shall be in accordance with MIL-G-23083. Dimensions shall be in accordance with MS17399.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. The contractor shall subject the system and all components to all the applicable tests of this specification. If all the components of the system are not procured from the same contractor, those components not procured shall be Government furnished to assure that the procured components meet the system requirements as well as the components requirements of this specification. When more than one set of Government furnished components is required for testing, the contractor shall provide the necessary simulated component signals.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval.

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4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include all the tests of 4.8.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the Acceptance Tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all Acceptance Tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance Tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

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(3) Reliability Assurance Tests

(4) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to individual tests. The tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1)	Examination of Product	4.8.1
(2)	Initial Erection	4.8.2
(3)	Final Erection	4.8.3
(4)	Power Consumption	4.8.4

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(5)	Erection Rate	4.8.5
(6)	Roll and Pitch Erection Cutout	4.8.6
(7)	Manual Fast Erection	4.8.7
(8)	Drift	4.8.8
(9)	Loop Rate	4.8.9
(10)	Power Interruption	4.8.10
(11)	Gimbal Freedom (Displacement Gyroscope)	4.8.11
(12)	Power Warning Circuit	4.8.12
(13)	Starting	4.8.13
(14)	Pitch and Bank Zero	4.8.14
(15)	Pitch Trim	4.8.15
(16)	Roll Adjustments	4.8.16
(17)	Follow-Up Accuracy	4.8.17
(18)	Sensitivity	4.8.18
(19)	Indication Lag	4.8.19
(20)	Hunting or Jumping	4.8.20
(21)	Overshoot	4.8.21
(22)	Power Warning Indicator	4.8.22
(23)	Fogging	4.8.23
(24)	Lighting	4.8.24
(25)	Insulation Resistance	4.8.25
(26)	Case Leakage	4.8.26
(27)	Vibration Error Gyroscope	4.8.4.1.1

* 4.3.2 Sampling Plans - The Sampling Plans shall consist of Sampling Plan A and Sampling Plan B. The tests samples selected for sampling tests shall first have passed the Individual Tests. The test samples which have been subjected to Sampling test shall not be delivered on contract until they have refurbished and resubmitted and passed all the Individual tests.

Quantity Offered for Acceptance	Quantity to be Selected for Sampling Plan A Tests
First 10	*1
Next 50	1
Each 100	1
of remainder	*Zero when Sampling Plan B is invoked

When an instrument failure occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of each fifteen from those still on hand, and proceed with sampling as indicated above when the next quantity is offered for acceptance. One additional sample, selected at random, shall be tested from each monthly lot for shelf life in accordance with 4.8.5.1. If the lot size is less than 25 instruments, the quantity produced in two months shall be considered as one lot. Concurrent delivery shall be permitted during the storage period.

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* 4.3.2.1 Scope of Sampling A Tests - Each system selected for Sampling Plan A Tests shall be subjected to the following tests:

(1)	Individual Tests	
(2)	Pitch and Roll Outputs	4.8.27
(3)	Synchro Transmitter Characterists	4.8.28
(4)	Synchro Zeroing and Sensings	4.8.29
(5)	Interference Control	4.8.30
(6)	Power and Frequency Variation	4.8.31
(7)	Horizon Line and Bank Index Null Position Shift	4.8.32
(8)	Measurement and Weight	4.8.33
(9)	Low Temperature Operation	4.8.34
(10)	High Temperature Operation	4.8.35
(11)	Magnetic Effect	4.8.36
(12)	Gimbal Clearance	4.8.37
(13)	Roll and Pitch	4.8.38
(14)	Verticality	4.8.39
(15)	Attitude Indication	4.8.40

* 4.3.2.2 Sampling Plan B Test - Three samples shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. Unless otherwise directed, these samples shall be forwarded at the contractor's expense to a government laboratory designated by the procuring activity. The sample shall be plainly identified by a durable tag, securely attached and marked with the following information:

(Name of Equipment)
Submitted by (Manufacturer's name, date) for production Acceptance Sampling Plan B Test of MIL-23081C in accordance with Contract/Order No. _____ _____
Manufacturer's Part No. _____

* 4.3.2.3 Sampling Plan B Approval - Approval of Sampling Plan B instruments shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material or performance defect made evident during this test shall be corrected by the Contractor to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of instruments under the contract to cease until proper corrective action is approved and accomplished.

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* 4.3.2.4 Sampling Plan B Inspection - Each sample selected for Sampling Plan B testing shall be subjected to the following tests:

(a)	Sampling Plan A Tests	
(b)	Vibration Failure	4.8.41.2
(c)	High Temperature Exposure	4.8.42
(d)	High Altitude-Low Temperature	4.8.43
(e)	Temperature Shock	4.8.44
(f)	Impact Shock	4.8.45
(g)	Acceleration	4.8.46
(h)	Salt Spray	4.8.47
(i)	Humidity	4.8.48
(j)	Fungus	4.8.49
(k)	Sand and Dust	4.8.50
(l)	Shelf Life	4.8.51

* 4.3.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by the Production Acceptance Phase shall be conducted. Classification of a failure shall be in accordance with MIL-STD-781 and AR-34.

* 4.3.3.1 Reliability Assurance Production Acceptance Phase - Reliability Assurance Production Acceptance Testing shall be conducted through a production burn-in. Each equipment produced but not submitted to the Qualification Test shall be operated for 150 operating hours burn-in, and the test data (operating hours and failures) shall be totaled and compared to the Accept/Reject Criteria of Test Plan II of MIL-STD-781. (Extend the lines as necessary to accomodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. Delivery of equipment shall not be held up to await a decision under the test plan. In event an Accept Decision is reached, the test duration of each equipment shall be reduced to 100 operating hours. At any time that the current total of test hours and test failures plotted on Test Plan IV curves (extended) show a Reject Situation, the Procuring Activity shall be notified. The Procuring Activity reserves the right to stop the acceptance of equipment at any time that a Reject Situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a rejection decision. For the Production Acceptance Phase, Test Level E shall be used, except that the temperature extremes shall be +55C and -44C, and vibration shall be 0.5G. Details of the burn-in tests shall be included in the test procedures.

* 4.3.3.2 Test Details - The test details, such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristic to be measured, special failure criteria, preventive maintenance to be allowed during the test etc., shall be part of the test procedures to be submitted to and approved by the procuring activity prior to the beginning of the Acceptance Test Phase of the Reliability Assurance Tests. The minimum performance characteristics to be measured shall be as specified in 6.8.

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4.3.4. Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment, and to assure adequate quality control. The equipment to be subjected to special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.3.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

* 4.3.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity.

* 4.4. Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Longevity Tests- Longevity Tests shall be performed on the equipment in accordance with Test Plan XXVIII of MIL-STD-781. The contractor shall furnish all samples and shall be responsible for accomplishing the Longevity Test. The test shall be of 2,000 hours duration and shall be conducted on equipments that have passed the Individual Tests. The Longevity Test shall be performed under the conditions specified in 4.5.1 and Table VI. The Longevity Test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Tests, Sampling Tests, Reliability Tests, or Special Tests may be selected for Longevity Tests.) When Reliability Tests are conducted, the Longevity Test shall be omitted if, during the Reliability Tests, a quantity of equipments equal to, or more than, that listed below receive at least 2000 hours each of test time.

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Quantity of Equipment Offered for Acceptance	Quantity to be Selected for Longevity Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The Longevity Test shall be conducted under the following simulated service conditions:

Temperature	Normal Room
Altitude	Normal ground (0-5000 ft)
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)

TABLE VI

Conditions	Test Conditions	Test (Hours)
	DISPLACEMENT GYROSCOPE	
1	3° (± 1-1/2°) Total Angle, Pitch, Roll, and Yaw Motion 5 to 7 oscillations per minute, also, reversing direction each minute.	600
2	15° (± 7-1/2°) Total Angle, Pitch, Roll, and Yaw Motion 5 to 7 oscillations per minute also, reversing direction each minute.	600
3	Vibration - 2.2g's ± 10% peak acceleration value measured at the equipment mounting points at any nonresonant frequency or combination of nonresonant frequencies between 20 and 60 Hz.	600
4	Tests conducted in accordance with this specification as specified by the procuring activity.	200

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TABLE VI (Continued)

Conditions	Test Conditions	Time (Hours)
	INDICATORS	
	The indicator shall be properly connected and operated such that both the standard test roll transmitter and standard test pitch transmitter are oscillated greater than ± 180° from their 180° dial reading. The transmitters shall be oscillated at a rate of 5 to 6 cycles per minute. The switch that connects indicator Pins A to J shall be momentarily opened every 30 minutes. The lighting circuits shall be continuously energized with 5 + 0.0-0.2 volts applied between indicator Pins K and L.	2000

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - Once each day during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity. The minimum performance characteristics to be measured shall be as specified in 6.9.

4.5.4 Test Data - The contractor shall keep the daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

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4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting Preproduction, reliability, acceptance and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of Test Procedures.

4.7 Test Conditions -

4.7.1 Atmospheric Conditions - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of 3.3.8.

4.7.2 Attitude - Except as specified below, the Vertical Gyroscope Indicator System shall be mounted as follows:

(1) The Displacement Gyroscope Assembly shall be mounted and mechanically secured to a platform which is level within 0.1 degree and is capable of 360 degree rotation in roll and pitch. The pitch and roll axis of the gyroscope shall be aligned to the pitch and roll axis of the platform.

(2) The Rate Switching Gyroscope shall be mounted in the normal or inverted position on a rate table.

(3) The Remote Attitude Indicator shall be tested in a normal upright position.

4.7.3 Input Power - Unless otherwise specified below, the units of the Vertical Gyroscope Indicator System shall be tested with the following power inputs:

(1) Displacement Gyroscope Assembly - 115/200 volts three phase 4 wire wye 400 Hz. The phase sequence shall be A, B, C.

(2) Remote Attitude Indicator - 115 VAC single phase 400 Hz.

(3) Rate Switching Gyroscope - 115 VAC single phase 400 Hz.

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4.7.4 Test Equipment and Connections - Whenever practical, the test equipment used in performing the tests shall not affect the performance of the equipment under test.

4.7.5 Interconnections - The components of the Vertical Gyroscope Indicator System shall be properly interconnected in accordance with Figure 1 or 2, as applicable.

4.7.6 Preliminary Adjustments - The following adjustments and control settings shall be made prior to testing of the Vertical Gyroscope Indicator System and shall remain at this setting, unless otherwise specified.

(1) Adjust the roll and pitch trim of the Attitude Indicator for zero trim.

4.8 Test Methods -

4.8.1 Examination of Product - Each equipment shall be examined externally to determine conformance to the applicable drawings and to all the requirements of this specification not covered by tests.

4.8.2 Initial Erection - The Displacement Gyroscope shall be leveled to within plus or minus six minutes. The spin axis of the Vertical Gyroscope under test shall be at any random position within the limits of plus or minus 85 degrees roll and plus or minus 82 degrees pitch of its operating vertical position. Power shall be applied to the Gyroscope. The Gyroscope shall erect to within plus or minus one degree in pitch and roll by the end of the initial erection time delay period. The time delay period shall not exceed 68 seconds as determined by a connection between Pins J and T for power flag excitation. This test shall be conducted without scorsby or vibration. At Low Temperature or Low Voltage and Frequency, the time delay period shall not exceed 90 seconds.

4.8.2.1 Gyroscope Motor Failure Monitor - At the end of the initial erection period the circuit between T and J shall immediately close indicating that the gyroscope motor is in excess of 5000 RPM.

4.8.3 Final Erection - The Displacement Gyroscope shall be allowed to erect for two minutes after the Initial Erection Test. The spin axis of the Gyroscope shall have aligned itself with the gravity of the vertical within 1/4 degree. The Gyroscope shall be rapidly rotated 180 degrees in azimuth. The pitch or roll axes shall not change by more than 1/2 degree. This test shall first be conducted without scorsby or vibration. Then on a low angle (not to exceed two degrees total movement) Roll, Pitch and Yaw mechanism operated at a speed of five to seven oscillations per minute.

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* 4.8.4 Power Consumption - The power consumption of the Vertical Gyroscope Indicator System will be measured and shall not exceed the following limits:

(1) Displacement Gyroscope

(a) Running 35VA
(b) Fast Erection or Starting 55VA

(2) 3-Inch Remote Attitude Indicator - 22VA

(3) 5-Inch Remote Attitude Indicator - 22VA

(4) Rate Switching Gyroscope - 10VA

4.8.5 Erection Rate - The Displacement Gyroscope shall be deflected 10 degrees in right roll and the time required for the instrument to erect from 10 degrees to five degrees recorded. The computed erection rate shall be between the limits of 1.0 degree and 3.0 degrees per minute. The same test shall be conducted for left roll, dive and climb, and the same tolerance shall apply. While erecting, the Gyroscope shall not deviate in the undeflected axis by more than one degree. This test shall be conducted with the Gyroscope mounted on a low angle (two degrees total movement) roll, pitch, and yaw mechanism.

4.8.6 Erection Cutout

4.8.6.1 Roll and Pitch Erection Cutout - The Displacement Gyroscope shall be deflected 10 degrees in roll and 10 degrees in pitch. Pin G shall be connected to Pins K and L. There shall be no evidence of erection in either the roll or the pitch axis.

4.8.6.2 Roll Erection Cutout, Internal - With the gyroscope in normal operation, the spin axis shall be precessed in bank. There shall be no evidence of erection in roll when the bank angle exceeds 7 $\pm$ 1 degree. Roll erection shall continue to be cutoff for a period of 5 $\pm$ 1 minutes when the bank angle is maintained in excess of the cutoff angle. With the bank angle maintained in excess of the cutoff angle roll erection shall begin to operate again within 5 $\pm$ 1 minute. Roll erection cutoff shall not be operative again until the gyroscope roll axis gets within 4 degrees of vertical. The test shall be performed in both left bank and right bank.

4.8.7 Manual Fast Erection - With the Displacement Gyroscope deflected, as specified in the roll and pitch erection cutout test, Pin A, B, or C shall be connected to Pin N. The Gyroscope shall immediately start to erect at a rate of at least 15 degrees per minute.

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* 4.8.8 Drift - The Displacement Gyroscope shall be mounted on a roll, pitch, and yaw mechanism, with the connector end of the Gyroscope pointed to the geographic west. The spin axis shall be deflected between 20 degrees from the vertical in an easterly direction and the erection disconnected. The roll, pitch, and yaw mechanism shall be operated at a speed of five to seven oscillations per minute and a total movement of 2-1/2 degrees. With proper correction for the earth's rotation, the computed drift rate in either axis shall not exceed 15 degrees per hour. The test shall be repeated with the gyroscope case pointed to the south, west, and north. The same tolerances shall apply. If the drift in one heading exceeds the limit by not more than one degree in five minutes, the Gyroscope may be retested in that heading and accepted if the first repeat test is in limits. Drift readings shall be recorded after a minimum period of five minutes on each heading.

* 4.8.9 Loop Rate - The Displacement Gyroscope shall be mounted on a fixture permitting rotation about a horizontal axis. The Gyroscope shall be rotated one complete revolution (loop) at a rate of 10 degrees plus or minus one degree per second and returned to the normal position. The error in either pitch or roll shall not exceed two degrees. The test shall be repeated for rotation about the other horizontal axis.

4.8.10 Power Interruption - The Displacement Gyroscope shall be properly connected and after five minutes of operation, the power shall be removed and the gyroscope moved through angles of plus or minus 60 degrees in pitch and then plus or minus 75 degrees in roll and returned to level. (Note: The intention is to suggest equal displacements plus and minus in pitch or roll.) Power shall be applied 30 seconds after removal. The pitch and roll axes shall indicate level within plus or minus one degree and there shall be no recycling of the initial erection. The circuit between Pins J and T shall immediately close.

* 4.8.11 Gimbal Freedom (Displacement Gyroscope) - With power applied to the Displacement Gyroscope, the Gyroscope shall be moved 360 degrees in roll and plus or minus 82 degrees in pitch. There shall be no indication of gimbal sticking or binding in either axis.

4.8.11.1 Roll Performance - Smoothly rotate the Displacement Gyroscope 360 degrees in the roll at a rate of 2 rpm (max.), 1.5 rpm (min.); the roll indication shall follow smoothly and shall not jump greater than one degree in roll or pitch. Perform this test for both directions of roll.

4.8.11.2 Pitch Performance - Smoothly rotate the Displacement Gyroscope 360 degrees in pitch at a rate of 2 rpm (max.), 1.5 rpm (min.); the pitch indication shall follow smoothly and shall not jump greater than one degree in roll or pitch except as specified below. Perform this test for both directions of pitch.

(1) At a minimum of 82 degrees climb and 82 degrees dive a 180 degree roll indication shall occur.

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*4.8.12 Power Warning Circuit - The Displacement Gyroscope shall be properly connected and power applied. The time required for the circuit between J and T to close shall not exceed 68 seconds. After the Gyroscope has reached full speed, the external connection to Pin A shall be removed, and the circuit between Pins J and T shall open immediately. The connection to Pin A shall be returned and the circuit betweens Pins J and T shall close. This procedure shall be repeated with Pin B and then Pin C. The same results shall occur as with Pin A. If at any time, any phase of the 3-phase power to the Gyroscope is removed, a warning voltage (VAC) shall be provided.

4.8.13 Starting - The Attitude Indicator shall be properly connected and when power is applied, the warning flag shall disappear and the Indicator shall be ready for immediate operation.

4.8.14 Pitch and Bank Zero - With the standard test transmitters for pitch and roll set at electrical zero and the pitch trim knob set to its zero position, the Indicator shall indicate zero pitch within 1/2 degree, and the roll adjustment shall be adjusted for a zero roll indication within 1/2 degree.

With the standard test transmitters for pitch and roll set to electrical zero, apply power and the spheroid shall follow-up to within 1/2 degree of zero in pitch and roll from any random position.

The pitch and roll transmitters shall be displaced to any random position, Indicator power removed, and the transmitters returned to electrical zero. When power is reapplied, the spheroid shall follow-up to within 1/4 degree of the preceding readings.

*4.8.15 Pitch Trim - With the standard test transmitters set to electrical zero, the pitch trim knob shall be rotated to its limits of one-quarter turn counterclockwise and one-half turn clockwise and the deflection of the horizon line observed. The respective deflections shall be from five degrees to 10 degrees climb and from 10 degrees to 20 degrees dive from the level flight indications. With the pitch trim knob in the zero pitch trim position, the Indicator shall indicate zero pitch within one-half of the horizon line width. The knob shall have sufficient friction to remain fixed when set, but it shall move smoothly without binding or erratic motion when being adjusted. Two Indicators shall be connected to the same transmitter and the pitch trim knob shall be moved through its range of motion on one of the Indicators. There shall not be more than 1/2 degree change in the indication of the other Indicator when a standard gyro synchro null voltage is provided.

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* 4.8.16 Roll Adjustments - The Attitude Indicator shall be properly connected to the roll standard test transmitter set to electrical zero. The roll zeroing adjustment shall be moved to its extremes and the deflection of the bank index noted. This deflection shall not be less than plus or minus three degrees from the zero position.

* 4.8.17 Follow-Up Accuracy - The rotor of the standard test transmitter in the roll axis shall be displaced to at least five random positions other than the zero position. The Indicator shall position itself with the rotor of the standard test transmitter within plus or minus 1/2 degree for each position between the 30 degrees left and right bank and within 1.0 degree for any position over the remaining 300 degrees. The rotor of the standard test transmitter in the pitch axis shall be displaced to at least five random positions other than the zero position. The Indicator shall position itself with the rotor of the standard test transmitter within plus or minus 1/2 degree for each position between 30 degrees dive and climb and within one degree for any position between 30 and 90 degrees plus or minus.

4.8.18 Sensitivity - The test transmitter for the roll axis shall be moved 1/4 degree in the right bank and returned to the original position, then 1/4 degree in the left bank and returned to the original position. The test transmitter for the pitch axis shall be moved 1/4 degree in dive and returned to the original position, then 1/4 degree in climb and returned to the original position. There shall be a smooth perceptible movement of the Indicator Bank Index and Horizon Line for each movement of the appropriate test transmitter. The test shall be repeated in dive and climb with the pitch trim in each of its extreme positions, except the test transmitter shall be moved 1/2 degree in each direction.

* 4.8.19 Indication Lag - The indication in the pitch axis (dive and climb) shall not lag behind the pitch remote input transmitter by more than five degrees for the shaft speed of 90 degrees per second. The indication in the roll axis (right and left) shall not lag behind the roll remote input transmitter by more than four degrees for the shaft speed of 90 degrees per second and 10 degrees for the shaft speed of 180 degrees per second.

* 4.8.20 Hunting or Jumping - There shall be no noticeable hunting or jumping of the Indicator Display in pitch or roll with no rate inputs from the test transmitters. The test transmitter shafts shall be rotated in both directions separately through all rates from zero through the maximum speed of the respective axis of display. The survey of rates shall be sufficiently slow to record the hunting and jumping that

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occur at any speed of each axis of display of the Indicator and shall not take less than one minute with the rate varying approximately linearly. The peak-to-peak amplitude of the hunting and jumping shall not exceed 1/2 degree for shaft speeds between zero degree and 20 degrees per second in pitch and roll, 1/2 degree for shaft speeds between 20 degrees and 90 degrees per second in pitch and roll, nor one degree for shaft speeds between 90 degrees and 300 degrees per second in roll.

* 4.8.21 Overshoot - Overshooting of the spheroid for both directions of pitch and roll shall not exceed an amplitude of three degrees and the spheroid shall come to rest within 0.5 seconds following a step input equal to the maximum follow-up lag.

* 4.8.22 Power Warning Indicator - The system shall be properly connected except that no power shall be applied. Power shall be applied and the flag shall disappear.

(1) For the Vertical Gyroscope Indicator System, loss (less than 70 V) of any single or combination of phases shall cause the flag to immediately appear.

(2) For the Vertical Gyroscope Indicator System the power failure flag will appear when the Gyro Wheel speed drops below 5000 RPM.

(3) For the Attitude Indicator, the power shall be disconnected and the flag shall immediately appear. Power shall be reapplied and the jumper from Pin J to Pin A shall be disconnected and the flag shall immediately appear.

4.8.23 Fogging - The Attitude Indicators shall be energized and placed in a 71 ± 2°C controlled ambient temperature for a minimum period of one hour. After this period, and while still at this temperature, an ice cube shall be rubbed on the Indicator cover glass for a period of from one to two minutes. The glass shall be wiped dry (do not use compressed air) and the Indicator inspected for evidence of water or oil fog inside the cover glass. Evidence of fogging shall be cause for rejection.

4.8.24 Lighting

4.8.24.1 Prime Standard Testing - Lighting tests of indicators to be used as a prime standard and production indicators submitted for acceptance shall be tested as specified in MIL-L-25467.

4.8.24.2 Production Testing - Lighting tests on production indicators shall be accomplished as follows:

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(1) Apply 5 ± 0.1 volts for Type I (3-Inch) or 28 volts for Type II (5-Inch) to the prime standard and to the production indicator to be checked. The two indicators shall be compared and shall visually appear identical in color.

(2) For the Type I (3-Inch), apply 3.0 ± 0.1 volts to the prime standard and 3.2 volts to the indicator being checked. For the Type II (5-Inch), apply 15 ± 0.1 volts to the prime standard and 16 volts to the indicator being checked. In this configuration the instrument being checked shall visually appear no dimmer than the prime standard.

(3) For the Type I (3-Inch), apply 3.0 ± 0.1 volts to the prime standard and 2.8 volts to the indicator being checked. For the Type II (5-Inch), apply 15 ± 0.1 volts to the prime standard and 14 volts to the indicator being checked. In this configuration the instrument being checked shall visually appear no brighter than the prime standard.

(4) An indicator which does not meet the visual comparison tests listed above shall not be approved unless the indicator is submitted to all display tests listed in MIL-L-25467 and sufficient data is taken to substantiate conformance to the display requirements of MIL-L-25467.

* 4.8.25 Insulation Resistance - A potential of 500 volts direct current shall be applied between isolated pins and between pins and the case of each component chassis for a period of five seconds. There shall be no breakdown of insulation and the resistance between the chassis and any circuit shall be not less than 50 megohms. No tests shall be made in circuits connected with the Vertical Gyroscope Erection Switches.

* 4.8.26 Case Leakage - Each component shall be tested for case leakage by a Mass Spectrometer Leak Test. The gas or mixture of gases in the case, adjusted to a pressure differential of one atmosphere, shall not have a leakage rate from the case exceeding one micron cubic foot per hour (10^{-5} cubic centimeters per second).

* 4.8.27 Pitch and Roll Output Gyroscope - The Gyroscope shall meet the requirement of Paragraph 3.5.1.7.

4.8.27.1 The 700 volts transient of 25 milliseconds duration shall be applied at Pins A, B, and C, simultaneously. At the conclusion of this test, the Gyroscope shall be subjected to and shall meet the requirements of the Individual Tests.

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4.8.28 Synchro Transmitter Characteristic - The Gyroscope shall meet the requirements of Paragraph 3.5.1.7.1.

4.8.29 Synchro Zeroing and Sensing, Gyroscope - The Gyroscope shall meet the requirements of Paragraph 3.5.1.8 and 3.5.1.8.1.

* 4.8.30 Interference Control - The Vertical Gyroscope Indicator System, properly connected, and operating, shall be tested in accordance with MIL-STD-461, Class ID Equipment.

During the conducted and radiated Interference Tests, the Remote Attitude Indicator shall be operated in pitch and roll. During susceptibility tests, the Indicator shall be operating but the pitch and roll axes shall be stationary at any random test transmitter setting. The Gyroscope shall be mounted on a Scorsby Table with a total movement of three degrees, five to seven oscillations per minute; reversing every minute.

4.8.31 Power and Frequency Variation -

* 4.8.31.1 The equipment shall be subjected to the variation of voltage and frequency of 4.8.31.1.1 and shall meet the following requirements at the four extreme conditions as a minimum:

(1) The Displacement Gyroscope shall meet the requirements of 4.8.2 (Initial Erection), 4.8.3 (Final Erection), 4.8.5 (Erection Rates), 4.8.8 (Drift), 4.8.11 (Gimbal Freedom), and 4.8.12 (Power Warning Circuit).

(2) The Attitude Indicator shall be subjected to and shall meet the requirements of 4.8.14 (Pitch and Bank Zero), except the 4.8.13 (starting) bank zero tolerance shall be 3/4 degree; 4.8.18 (Sensitivity), except the tolerance shall be one degree in roll and 3/4 degree in pitch; 4.8.11 (Gimbal Freedom); and 4.8.22 (Power Warning Indicator).

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4.8.31.1.1

A. C. Volts	Frequency Hz
104 to 122	360 to 440

4.8.31.2 The equipment shall be run for 10 minutes at each condition of voltage and frequency as shown below:

A. C. Volts	Frequency Hz
102	370 and 430
124	370 and 430

During these conditions, the equipment:

(1) May have momentary loss of function; however, this momentary loss shall not effect the equipment performance later.

(2) After the equipments return to normal operation, it shall recover automatically to specified performance. At the conclusion of this test, the equipments shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.32 Horizon Line and Bank Index Null Position Shift - The Indicator shall be properly connected to standard test transmitters in the pitch and roll axes. The transmitter shall be set to drive the indicator in both axes to the zero position. The output voltage and frequency shall be varied within the limits and combinations listed below and the indicator shall meet the following requirements:

(1) The horizon line shall not shift more than plus or minus 1/32 inch from normal null position.

(2) The bank index shall not shift more than one-half of the narrow end of the index width from normal null position. At the conclusion of this test, the Indicators shall be subjected to and shall meet the requirements of the Individual Tests.

A.C. Volts	Frequency Hz
102 to 124	420
102 to 124	380
102	380 to 420
124	380 to 420

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* 4.8.33 Measurement and Weight - The components of the system shall be weighed and the case dimensions measured. Weights and measurements of each component shall conform to the following requirements:

(1) Displacement Gyroscope -

(a) The outline case dimensions shall conform to MS14114-1.

(b) The weight shall not exceed 10.5 pounds with adapter plate.

(2) 3-Inch Remote Attitude Indicator -

(a) The outline case dimensions shall conform to MS17400.

(b) The weight shall not exceed 3.8 pounds.

(3) 5-Inch Remote Attitude Indicator -

(a) The outline case dimensions shall conform to MS17401.

(b) The weight shall not exceed 5.0 pounds.

* 4.8.34 Low Temperature Operation - The equipment shall be properly connected, except that no power shall be applied. The equipment shall be tested in accordance with Step 2, Temperature and Altitude Test, of MIL-T-5422, (except as modified below) for four hours. At the end of this four-hour period and while at this reduced temperature, power shall be applied. The equipment shall be subjected to and shall meet the requirements of the following:

* 4.8.34.1 Displacement Gyroscope - The Tests 4.8.2 (Initial Erection), 4.8.3 (Final Erection), 4.8.4 (Power Consumption), 4.8.5 (Erection Rates), 4.8.7 (Manual Fast Erection), 4.8.8 (Drift), 4.8.11 (Gimbal Freedom), and 4.8.13 (Power Warning Circuit) shall be performed. The Displacement Gyroscope shall meet the requirements specified for these tests with the following exception:

4.8.34.1.1 Before conducting the Initial Erection Test, the spin axis shall have been displaced at any random pitch angle within plus or minus 82 degrees and any random roll angle within plus or minus 80 degrees from the vertical at the time power is applied and the initial erection time shall not exceed 1-1/2 minutes after power is applied.

4.8.34.1.2 The erection rate limits shall be 1.0 to 3.0 degrees per minute with a one-degree maximum deviation on the undeflected axis.

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4.8.34.1.3 The no-load output shall be 11.8 ± 0.4 V and the no-load phase shift shall not exceed 5.5 degrees. The line-to-line output voltage when loaded with five loads as specified in 3.5.1.7.1 shall not be less than 10.7V. The phase shift with five loads shall not exceed 9 degrees.

4.8.34.2 Attitude Indicator - The Test 4.8.15 (Pitch and Bank Zero) shall be performed, except the tolerance shall be plus or minus one degree, and Test 4.8.19 (Sensitivity) shall be performed, except that the tolerance in pitch and roll (with the pitch trim knob in the zero position) shall be plus or minus 1/2 degree and the tolerance in dive and climb (with the pitch trim knob in each of its extreme positions) shall be plus or minus one degree.

4.8.34.3 Vertical Gyroscope Indicator System - The Tests 4.8.39 through 4.8.42 shall be performed. The Erection Rate (4.8.40) shall be 1.0 to 3.0 degrees per minute and the Attitude Indication (4.8.41) shall not change more than 0.65 degree.

4.8.34.4 At the conclusion of this test and after the equipment has returned to room temperature, the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.35 High Temperature Operation - The equipment shall be properly connected and tested in accordance with Steps 6 and 7, Temperature and Altitude Tests, of MIL-T-5422. At the end of each step and while still at the elevated temperature, the equipment shall be subjected to and shall meet the requirements of the following:

4.8.35.1 Displacement Gyroscope - The Tests 4.8.2 (Initial Erection), 4.8.3 (Final Erection), 4.8.4 (Power Consumption), 4.8.5 (Erection Rates), 4.8.7 (Manual Fast Erection), 4.8.8 (Drift), 4.8.11 (Gimbal Freedom), 4.8.13 (Power Warning), and the Synchro Transmitter Characteristics Condition shall be as follows:

4.8.35.1.1 The no-load output shall be 11.8 ± 0.4V and the no-load phase shift shall not exceed 8 degrees. The line-to-line output voltage when loaded with five loads as specified in 3.5.1.7.1 shall not be less than 10.6 V. The phase shift with five loads shall not exceed 11.5 degrees.

4.8.35.2 Attitude Indicator - The Test 4.8.14 (Pitch and Bank Zero) shall be performed, except the tolerance shall be ± 1 degree, and Test 4.8.18 (Sensitivity) shall be performed, except the zero pitch trim tolerance shall be ± 1/2 degree and ± 1 degree with the pitch trim at the extremes.

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4.8.35.3 Vertical Gyroscope Indicator System - Upon completion of Step 6 of MIL-T-5422, the equipment shall be immediately subjected to Step 7 of MIL-T-5422. At the end of the 30 minute period and while still at the elevated temperature the equipment shall be subjected to and shall meet the requirements of the following:

4.8.35.3.1 For the Displacement Gyroscope: Tests 4.8.39 (Verticality), 4.8.40 (Manual Fast Erection), and 4.8.41 (Attitude Indication.

4.8.35.4 At the conclusion of this test and after the equipment has returned to room temperature, it shall be subjected to and shall meet the requirements of the Individual Tests.

4.8.36 Magnetic Effect - The Attitude Indicator shall be properly connected and power applied. The Indicator shall be revolved about a short bar magnet compass with the nearest part of the Indicator 5-1/2 inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The Indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar magnet. The Indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees. At each of these positions, the Indicator shall be rotated 360 degrees about its horizontal axis. The deflection of the compass at any of the specified positions shall not exceed one degree. This test shall be repeated with no power applied. The same tolerance shall apply.

4.8.37 Gimbal Clearance - The Displacement Gyroscope which has passed the Gimbal Freedom Test shall be subjected to a differential pressure equivalent to the differential between sea level and 70,000 feet. While exposed to this differential pressure, the Displacement Gyroscope shall meet the requirements of Gimbal Freedom (4.8.11). At the conclusion of this test, the Gyroscope shall meet the requirements of the Individual Tests.

* 4.8.38 Roll and Pitch -

4.8.38.1 Roll - The Displacement Gyroscope shall be mounted on a fixture rotating about a lateral axis. The Gyroscopes shall be rotated five complete revolutions (rolls) at a rate of 400 degrees plus or minus 10 degrees per second. The error induced in either pitch or roll shall not exceed one degree.

4.8.38.2 Pitch - The Displacement Gyroscope shall be mounted on a fixture Rotating about a Longitudinal Axis. The Gyroscope shall be rotated two complete revolutions (Dive then Climb) at a rate of 90 degrees per second. No parts of the Gyroscope shall be damaged by this rate. At the conclusion of this test, the Gyroscope shall be subjected to and shall meet the requirements of the Individual Tests.

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4.8.39 Verticality - Rotate the Displacement Gyroscope 180 degrees in azimuth; the roll and pitch indication of the Attitude Indicator shall not change more than 0.5 degree.

4.8.40 Attitude Indication - The roll and pitch performance of the Vertical Gyroscope Indicator System shall be tested by rotating the Gyroscope Assembly in roll and pitch as specified below:

4.8.40.1 Roll Performance - Smoothly rotate the Displacement Gyroscope 360 degrees in roll at a rate of 2 RPM (max.), 1.5 RPM (min.), the roll indication of the Attitude Indicator shall follow smoothly and the Indicator presentation shall not jump greater than one degree in roll or pitch. Perform this test for both directions of roll.

4.8.40.2 Pitch Performance - Smoothly rotate the Displacement Gyroscope 360 degrees in pitch at a rate of 2 RPM (Max.), 1.5 RPM (Min.), the pitch indication of the Attitude Indicator shall follow smoothly and the Indicator presentation shall not jump greater than one degree in roll or pitch, except as specified below. Perform this test for both directions of pitch.

(1) At a minimum of 82 degrees climb and 82 degrees dive the outer gimbal shall roll 180 degrees.

4.8.40.3 Servo Sensitivity - The Servo Sensitivity shall be tested with the Displacement Gyroscope at zero degrees roll and pitch. Displace the Displacement Gyroscope in roll 0.5 degree in both directions from zero; there shall be a perceptible movement of the attitude indication. Repeat this test for the pitch axis; the same results shall be obtained.

4.8.41 Vibration - The Vertical Gyroscope and the Attitude Indicator shall be exposed individually to the Vibration Tests as described in the following paragraphs:

4.8.41.1 Vibration Error -

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4.8.41.1.1 Displacement Gyroscope - The Displacement Gyroscope shall be subjected to a 20 minute scanned vibration resonance test in the longitudinal, lateral and vertical axes. A logarithmic rate of frequency change shall be used from 5 Hz to 500 and back to 5 Hz. Power shall be applied and the case displaced one degree in roll and pitch for resonance search and dwell modes. The vibration level from 5 Hz to 20 Hz shall be 0.1 inches double amplitude. The vibration level from 20 Hz to 500 Hz shall be 2 gs. The signal outputs (pitch and roll) shall be recorded on a strip chart recorder as a permanent record. A five minute dwell in each axis, at the most severe resonances frequencies, where the amplitude exceeds $1/2^{o}$, shall be made at the applicable vibration level. If more than three significant resonances are found, then the three most severe shall be chosen with at least two of the most severe being selected within the frequency range of 10 Hz to 35 Hz. If the level exceeds 1/2 degree and a predominant resonance cannot be determined then three frequencies separated by at least 7 Hz within the frequency range of 10 Hz to 35 Hz shall be selected for the dwell requirements. Production equipment subjected to the requirements of paragraph 4.3.6 individual tests shall be subjected to a five minute dwell in each axis at one of the most significant resonant frequencies between 10 Hz and 35 Hz determined during Preproduction testing. The pitch and roll outputs shall be monitored 100% of the time. During any five minute dwell failure to meet the following requirements shall be cause for rejection.

(1) The Pitch and Roll outputs shall not change more than + or -0.5 degrees from the starting point during the 5 minute dwell in each axis.

(2) There shall be no indication of output signal discontinuities or gimbal binding in either pitch or roll.

(3) The modulation of the output signals shall be less than 15% of the signal at 1° displacement.

(4) The synchro outputs in both axes shall change less than 0.5 degree during the 15 minute vibration period. Change is determined by placing the gyroscope on a table leveled within $\pm$ 6 minutes of arc before and after the vibration test.

NOTE: Starting point is defined as the recorded pitch and roll reading taken immediately prior to application of the specified Vibration level, with the gyroscope mounted on the vibrator and the case displaced one degree in roll and pitch.

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4.8.41.1.2 Remote Attitude Indicator - The Remote Attitude Indicator with power applied shall be vibrated 20 minutes in accordance with Curve II of Figure 2, of MIL-T-5422 except the 2g's portion of the curve shall be extended to 2,000 Hertz. The roll and pitch indication shall be observed during this vibration, and shall not hunt, jump, or oscillate greater than 1/2 degree. At the end of the 20 minutes vibration period, the pitch and roll axes shall not be more than 1/2 degree from their original position before vibration. This test shall be repeated with the vibration applied to each of the other two mutually perpendicular axes.

4.8.41.2 Vibration Failure -

4.8.41.2.1 Displacement Gyroscope - The Displacement Gyroscope shall be tested in accordance with Part I, Curve IV of MIL-T-5422, except the curve shall be extended to 2,000 Hertz. There shall be no signal transients or discontinuities in the pitch and roll synchro output circuits during the Vibration Test. At the conclusion of these tests, the Gyroscope shall be subjected to and shall meet the requirements of the Individual Tests. No damage or irregular operations shall result from the Vibration Tests.

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4.8.41.2.2 Attitude Indicator - The Attitude Indicator shall be tested in accordance with Part I, Curve III, of MIL-T-5422, except Curve III shall be extended to 2,000 Hertz. At the conclusion of these tests, the equipments shall be subjected to and shall meet the requirements of the Individual Tests. No damage or irregular operations shall result from the Vibration Tests.

4.8.42 High Temperature Exposure - The Vertical Gyroscope Indicator System shall be tested in accordance with Step 6, Temperature and Altitude Tests, of MIL-T-5422, except as modified for a period of 24 hours. The system shall be properly connected but no power shall be applied during the first 24 hours of exposure. At the end of the 24-hour period and while still at the elevated temperature, the equipment shall be subjected to and shall meet the requirements of 4.8.2 (Initial Erection), 4.8.3 (Final Erection), 4.8.4 (Power Consumption), 4.8.5 (Erection Rates), 4.8.8 (Drift), 4.8.11 (Gimbal Freedom), and 4.8.13 (Power Warning Circuit). Upon returning to room temperature, the system shall meet the requirements of the Individual Tests.

* 4.8.43 High Altitude-Low Temperature - The Vertical Gyroscope Indicator System, properly connected but with no power applied, shall be subjected to the following steps of the Temperature and Altitude Tests of MIL-T-5422, except as modified: Step 1 for 72 hours and Step 2 for 72 hours. After exposure to -54°C (-65°F) for 72 hours, the pressure in the chamber shall be reduced to the equivalent of 100,000 + 500 feet, power shall be applied, and the equipment shall meet the following requirements:

4.8.43.1 Displacement Gyroscope - The Tests 4.8.2 (Initial Erection), 4.8.3 (Final Erection), 4.8.4 (Power Consumption), 4.8.5 (Erection Rate), 4.8.6 (Roll and Pitch Erection Cutout), 4.8.7 (Manual Fast Erection), 4.8.10 (Power Interruption), and 4.8.11 (Gimbal Freedom) shall be performed.

4.8.43.2 Attitude Indicator - The Tests 4.8.20 (Indicator Lag) and 4.8.23 (Power Warning Indicator) shall be performed.

4.8.43.3 Vertical Gyroscope Indicator System - The Tests 4.8.41 through 4.8.43 shall be performed. The Erection Rate (4.8.42) shall be 1.0 to 3.0 degrees per minute and the Attitude Indication (4.8.43) shall not change more than 0.65 degree.

4.8.43.4 At the conclusion of these tests, the equipment shall be returned to room temperature and pressure and shall then be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.44 Temperature Shock - The Vertical Gyroscope Indicator System shall be subjected to the Temperature Shock Test, Procedure I, of MIL-E-5272, except the temperature range shall be -54°C to +95°C. At the conclusion of this test, the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

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* 4.8.45 Impact Shock - The Vertical Gyroscope Indicator System shall be tested in accordance with the Shock Test of MIL-T-5422, with the equipment non-operating and the gyroscope fixed to the mounting base for the 15 and 30 gs shock test.

4.8.46 Acceleration -

* 4.8.46.1 Acceleration (Nonoperating) - The Vertical Gyroscope Indicator System (Nonoperating) shall be subjected to 20g acceleration for a period of 1.0 minute in both directions along each of its three mutually perpendicular axes, while not operating. At the conclusion of the test, the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

4.8.46.1.2 Acceleration (Operating) - With the Attitude Indicator properly connected and power applied, the Indicator shall be accelerated along its normal vertical axis with a force of 5g for one minute. The attitude display shall not vary from the initial indication by more than 1.0 degree in both pitch and roll. This test shall be conducted with the attitude set at zero pitch and roll, with the roll axis displaced 90 degrees left and right bank and with the pitch axis displaced 80 degrees dive and climb.

* 4.8.46.2 Longitudinal Accelerations - The equipment shall meet the requirements of 3.4.1.5.

* 4.8.46.3 Coordinated Turns - The equipment shall meet the requirements of 3.4.1.6.

* 4.8.46.4 Uncoordinated Turns - The equipment shall meet the requirements of 3.4.1.7 (by flight test by the procuring agency if required).

4.8.47 Salt Spray - The Vertical Gyroscope Indicator System shall be tested in accordance with the Salt Spray Test of MIL-T-5422, except that the test period shall be 50 hours. At the end of the 50-hour period, the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.48 Humidity - The Gyroscope and Indicators shall be tested in accordance with the Humidity Test of MIL-T-5422, except 71°C shall be used in lieu of 50°C in Steps 1 and 2. At the completion of the test, the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

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* 4.8.49 Fungus Resistance - The Gyroscope and Indicators shall be tested in accordance with the Fungus Resistance Test of MIL-T-5422. After the test, any evidence of the growth of fungi found upon inspection shall be cause for rejection of the equipment or of lots represented. After completion of the test, the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

* 4.8.50 Sand and Dust - The Vertical Gyroscope and Indicators shall be tested in accordance with the Sand and Dust Test of MIL-T-5422. At the end of this test, the equipment shall be subjected to and meet the requirements of the individual tests.

* 4.8.51 Shelf Life - The instrument shall be placed in storage for a period of not less than 30 days. During the 30 day period, the instrument shall remain dormant as practical. The temperature shall remain constant; however normal room temperature variations will be permitted. At the end of the 30 day period, the instrument shall be subjected to and meet the Individual Test of paragraph 4.4.1.

4.8.52 Internal Examination - The case of the instrument be opened and the internal mechanism examined for compliance with the requirements of the contract, detail specification and all applicable specifications. Any deterioration or damage which could in any manner prevent the instrument from meeting functional operation or maintenance requirements during service life shall be cause for rejection.

4.9 Reconditioning of Test Equipment - Equipment which has been subjected to Acceptance and Longevity Tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, contractor shall resubmit the equipment for acceptance.

4.10 Presubmission Testing - No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.11 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector.

* 4.12 Maintainability Demonstration - Maintainability demonstration shall be in accordance with AR-10. Maintainability demonstration shall be performed on a minimum of three selected repairs. The tasks selected to demonstrate maintainability shall be selected in advance of timing the tasks and shall be approved by the procuring activity.

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5. PREPARATION FOR DELIVERY

* 5.1 Packaging, Packing, Preservation, and Marking - Packaging, Packing, Preservation, and Marking shall be in accordance with MIL-G-81559 requiring plastic fiber glass materials.

* 5.2 Reuseable Containers - Decals conforming to MIL-N-18307 shall be provided on each equipment except rate switching gyroscope, as follows:

NOTICE

"It is mandatory that upon return for repair or overhaul that this equipment be returned in reuseable container FSN _____ at time of equipment exchange.

Federal Stock Numbers shall be assigned to the applicable containers which shall be provided for initial or reordered equipment until an adequate supply level is attained for recycling into production or overhaul activities.

6. NOTES

6.1 Intended Use - The Vertical Gyroscope Indicator System covered by this specification is intended for use in high performance aircraft to provide continuous flight reference data.

6.2 Ordering Data - Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) The quantity, MS part number and type number, if any, of the instrument desired

(c) Levels of packaging and packing desired

(d) The laboratory that shall conduct test

(e) Reliability test phases (See 4.3)

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance, or reliability will result from the use of materials, parts, and processes other than those specified in MIL-E-5400; it is desired that their use be investigated. When investigation shows that advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification or from subsidiary specifications, where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications, referenced therein. All referenced specifications shall apply to the extent specified.

*6.5 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Monthly Reliability Reports in accordance with MIL-STD-781.

* 6.6 Asterisk - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in the notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.7 Items of Equipment - The equipment covered by this specification shall consist of the following items:

Type I	Type II
Displacement Gyroscope MS 14114	Displacement Gyroscope MS 14114
Attitude Indicator MS17400	Attitude Indicator MS17401
Rate Switching Gyroscope MS17399	Rate Switching Gyroscope MS17399

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* 6.8 Performance Characteristics to be Measured - At least once each day, the Vertical Gyroscope Indicator System and its associated components shall be subjected to and shall meet the requirements of the following tests in addition to a visual inspection for failure:

(1) Displacement Gyroscope -

(a) Initial Erection
(b) Final Erection
(c) Erection Rate
(d) Drift

(2) Attitude Indicator -

(a) Zeroing Adjustments
(b) Pitch and Bank Zero
(c) Sensitivity
(d) Gimbal Freedom

* 6.9 Supersession Data - This specification supersedes Military Specification MIL-G-23081B(AS) of 13 January 1967.

6.10 Definitions -

6.10.1 Hermetic Seal - A Hermetic Seal shall be a perfectly closed and airtight seal made between metallic and metallic or between metallic and vitric materials. A Hermetic Seal is not intended to include seals which are dependent upon gaskets.

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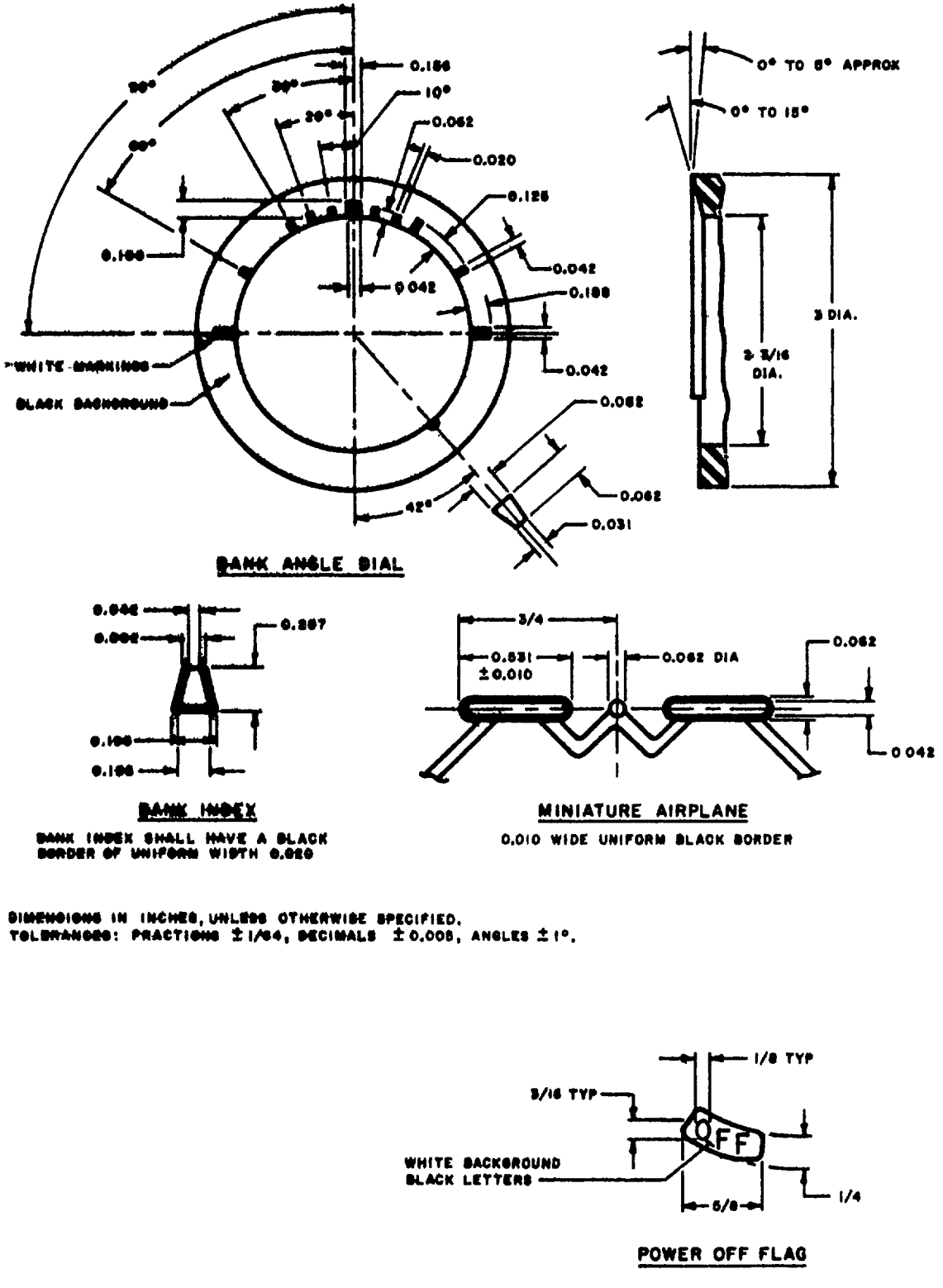
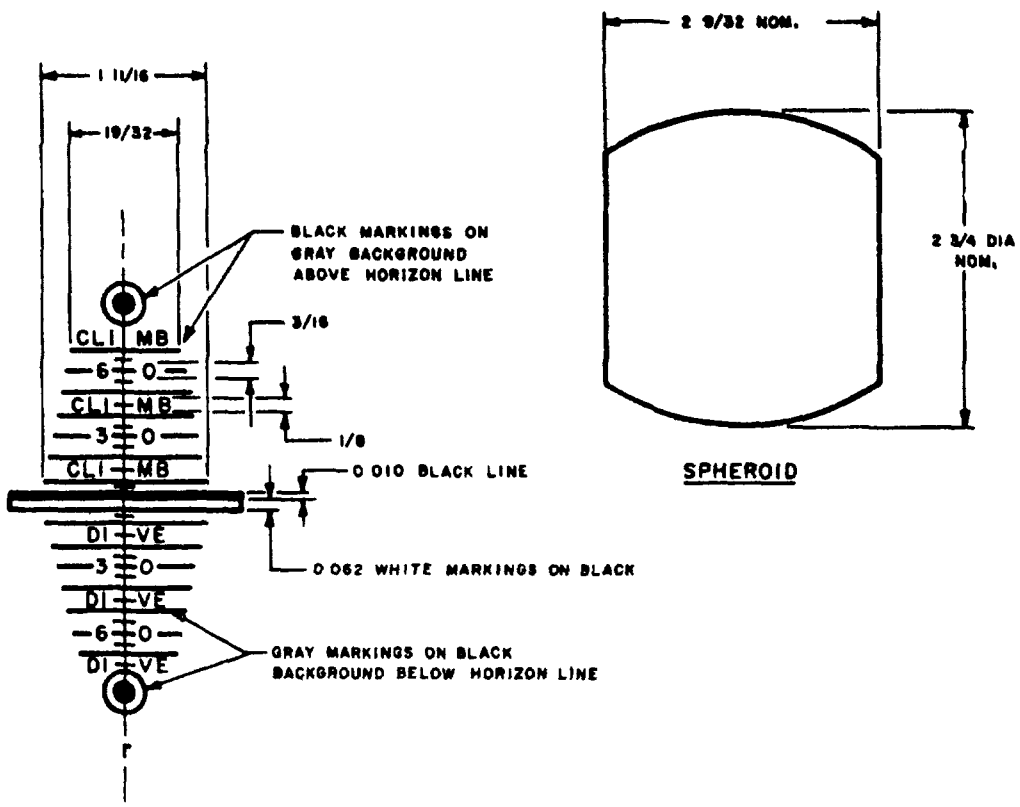


FIGURE 4. INDICATOR DISPLAY, TYPE I (3-INCH)

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Pitch indices which appear between pitch angle numerals shall be 7/32 inch long.

The pitch reference lines shall be placed on the spheroid so that they will appear straight when directly behind the miniature airplane.

All markings shall be 0.020 wide, unless otherwise specified.

Dimensions in inches. Unless otherwise specified, tolerances Fractions ±1/64, Decimals ±0.005.

The color and size of the markings shall be as specified herein.

FIGURE 5. SPHEROID AND MARKINGS, ATTITUDE INDICATOR, TYPE I (3-INCH)

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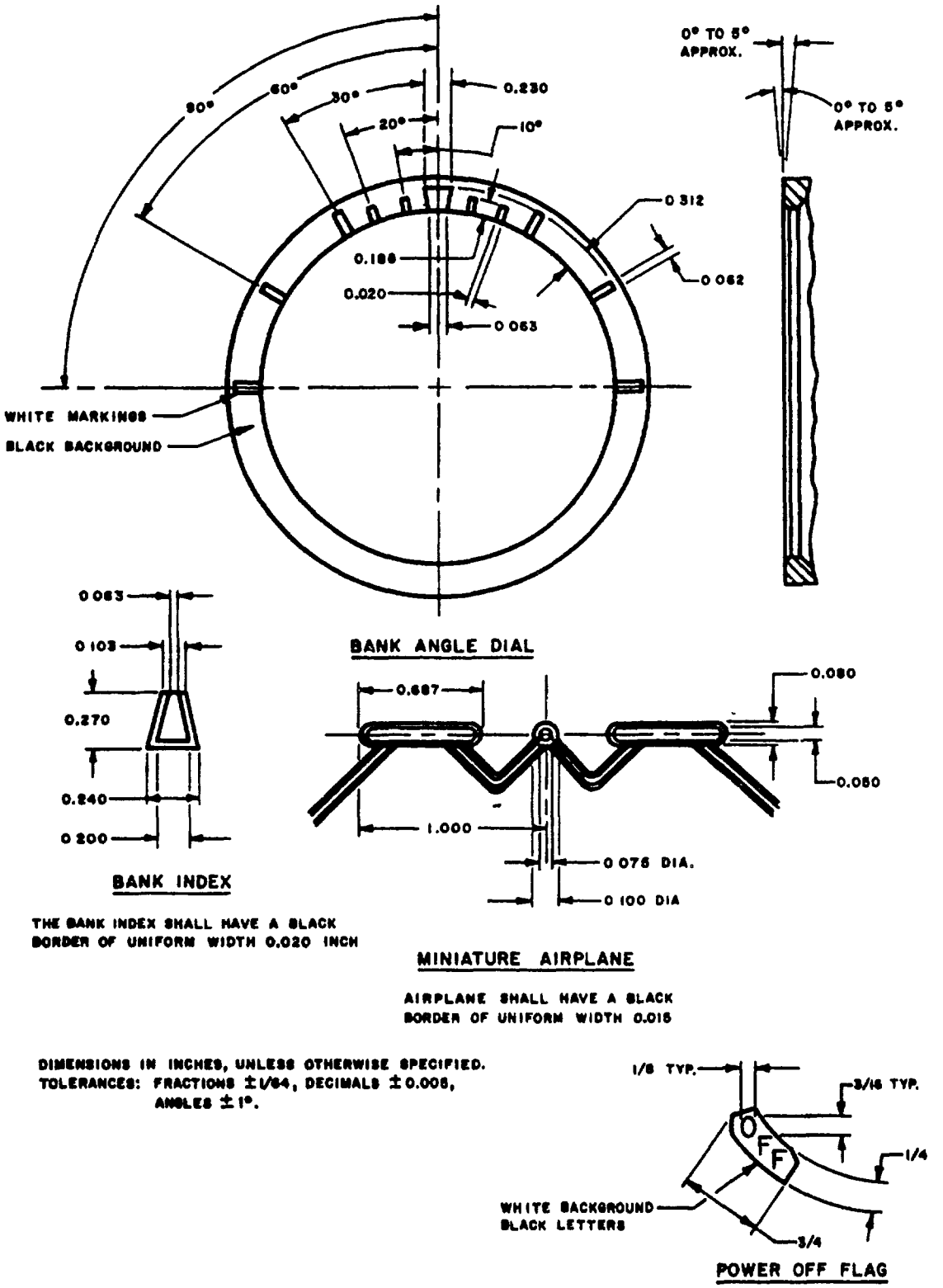
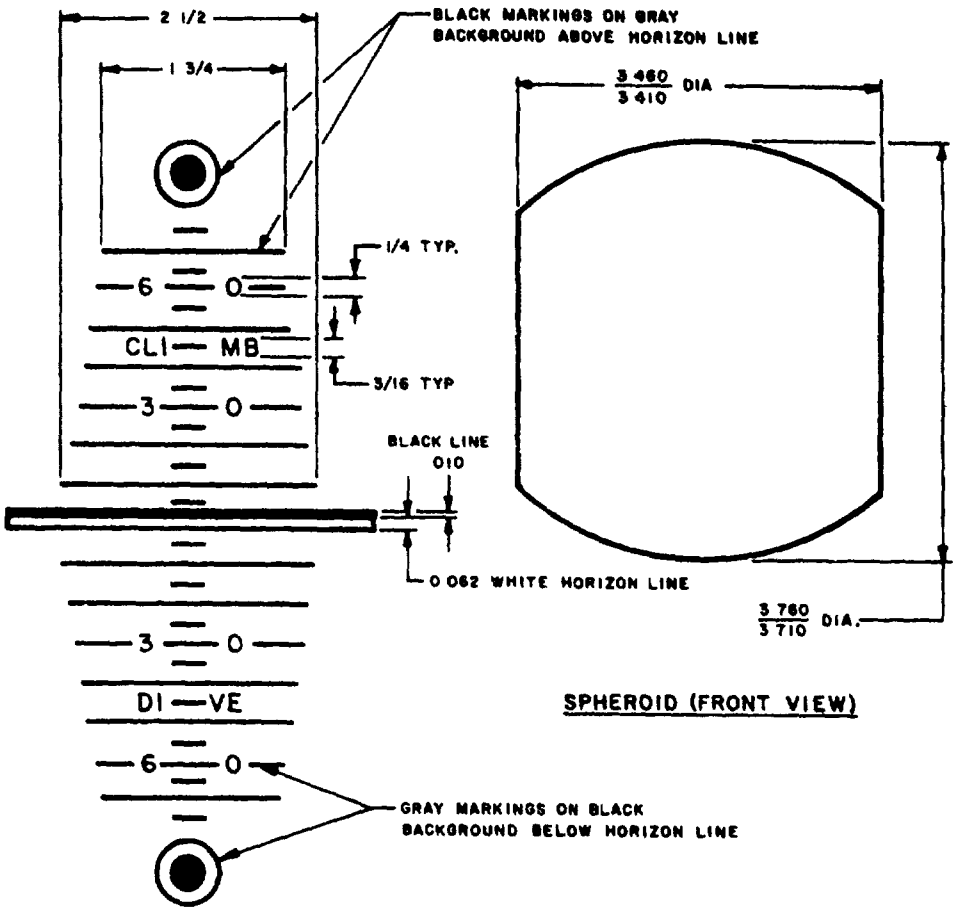


FIGURE 6. INDICATOR DISPLAY, TYPE II (5-INCH)

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The pitch reference lines shall be placed on the spheroid so that they are portions of great circles which intersect at the pitch axis.

All markings shall be 0.020 wide, unless otherwise specified.

Dimensions in inches. Unless otherwise specified, tolerances Fractions ±1/64, decimals ±0.005.

The color and size of the markings shall be as specified herein.

FIGURE 7. SPHEROID AND MARKINGS, ATTITUDE INDICATOR, TYPE II (5-INCH)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-G-23081C(AS) GYROSCOPE INDICATOR SYSTEM VERTICAL

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A GIVE PARAGRAPH NUMBER AND WORDING.

   B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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